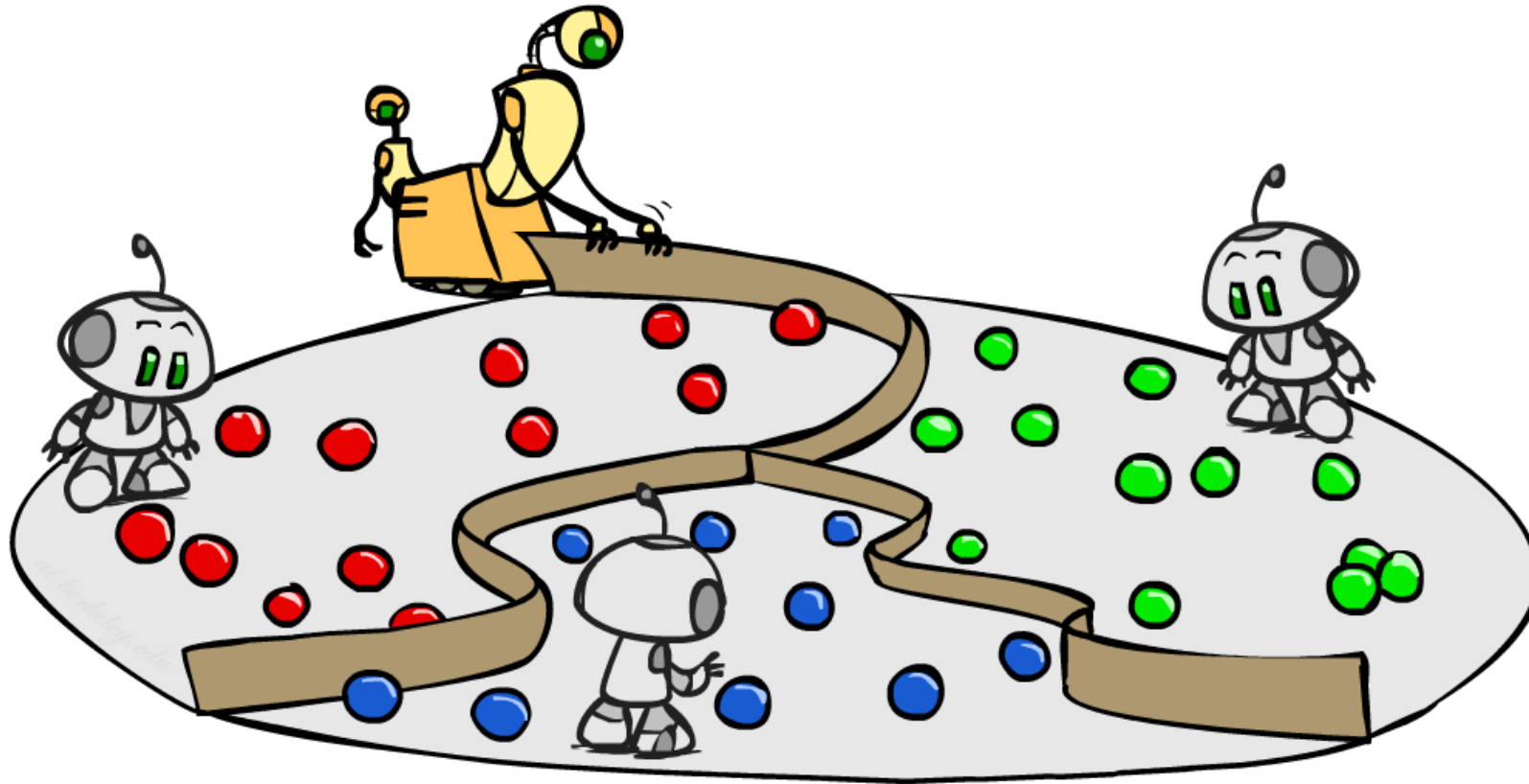


Project Presentation

- Project presentation
 - Week 16, in class

Unsupervised Machine Learning



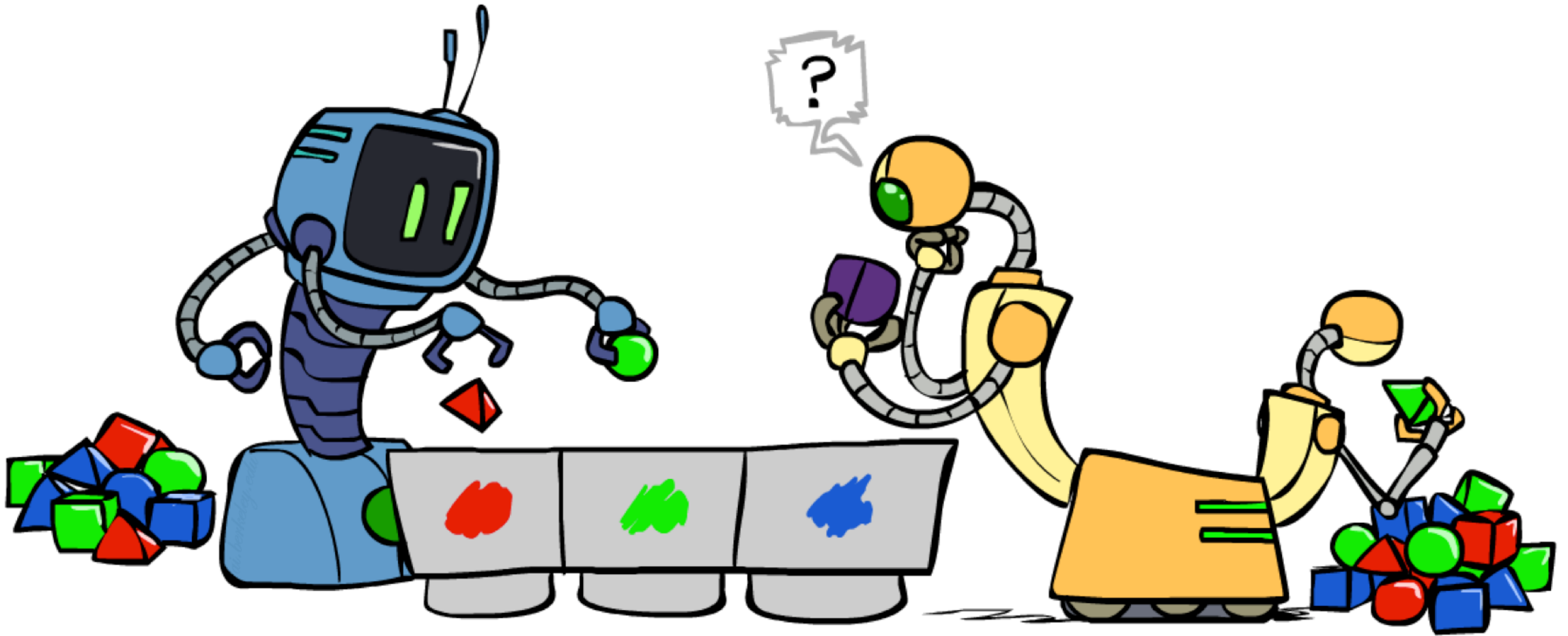
AIMA Chapter 20

[Adapted from slides by Dan Klein and Pieter Abbeel at UC Berkeley and from Daniel Weld, Carlos Guestrin, & Luke Zettlemoyer at U Washington]

Types of Learning

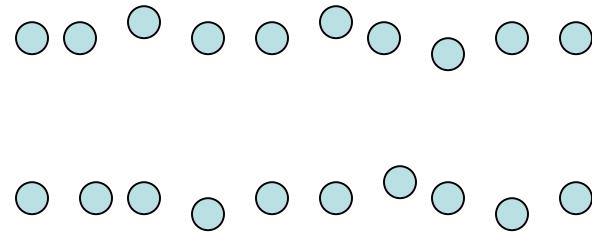
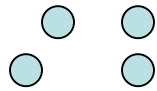
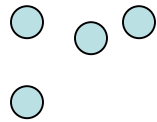
- Supervised learning
 - Training data includes desired outputs
- Unsupervised learning 
 - Training data does not include desired outputs
- Semi-supervised learning
 - Training data includes a few desired outputs
- Reinforcement learning
 - Rewards from sequence of actions

Clustering



Clustering

- Basic idea: group together similar instances
- Example: 2D point patterns



- What could “similar” mean?
 - One option: small (squared) Euclidean distance

$$\text{dist}(x, y) = (x - y)^{\top} (x - y) = \sum_i (x_i - y_i)^2$$

- Many other options, often domain specific

Clustering

- Applications
 - Group emails
 - Group search results
 - Find categories of customers
 - Detect anomalous program executions

Story groupings:
unsupervised clustering



World » [edit](#)

[Heavy Fighting Continues As Pakistan Army Battles Taliban](#)
Voice of America - 10 hours ago
By Barry Newhouse Pakistan's military said its forces have killed 55 to 60 Taliban militants in the last 24 hours in heavy fighting in Taliban-held areas of the northwest.
[Pakistani troops battle Taliban militants for fourth day](#) guardian.co.uk
[Army: 55 militants killed in Pakistan fighting](#) The Associated Press
[Christian Science Monitor](#) - [CNN International](#) - [Bloomberg](#) - [New York Times](#)
[all 3,824 news articles »](#)

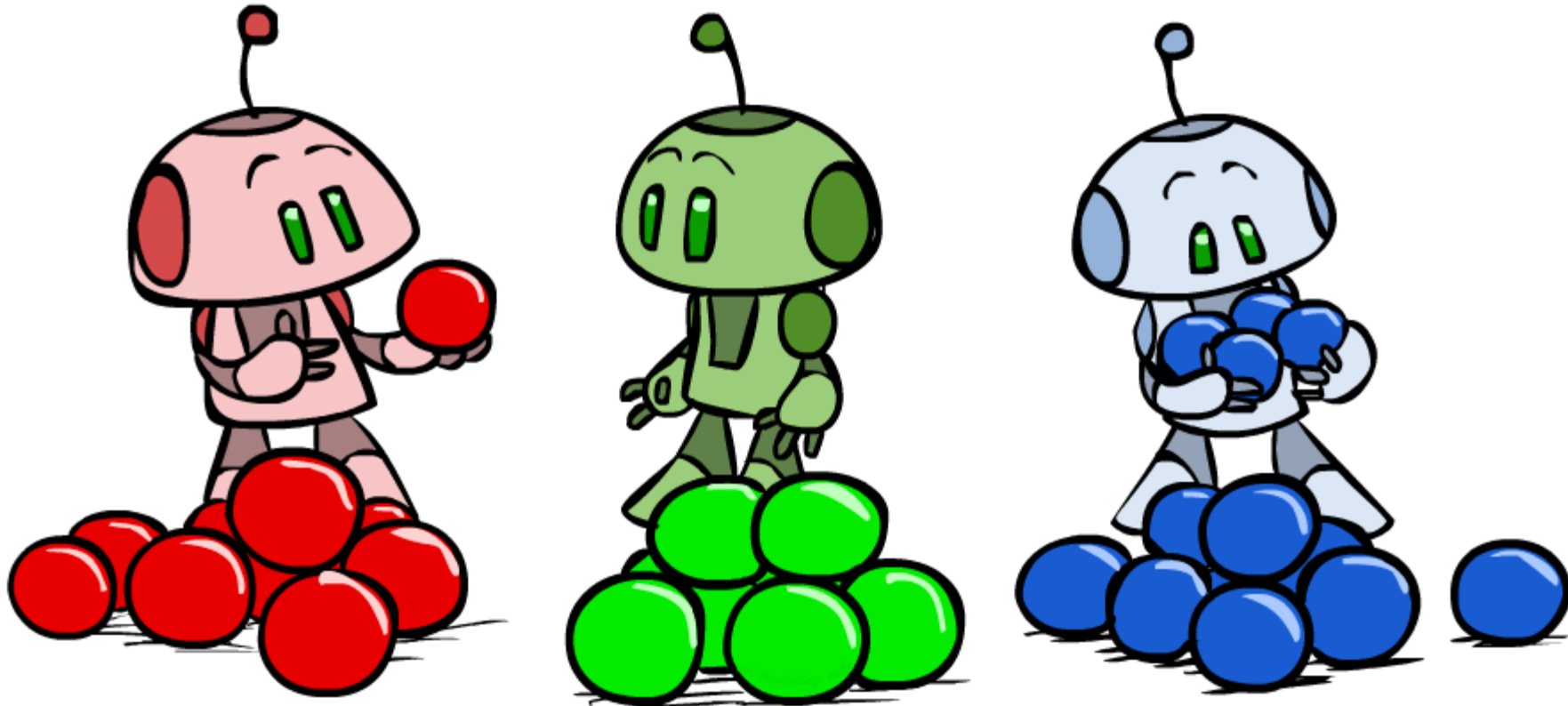
[Sri Lanka admits bombing safe haven](#)
guardian.co.uk - 3 hours ago
Sri Lanka has admitted bombing a "safe haven" created for up to 150000 civilians fleeing fighting between Tamil Tiger fighters and the army.
[Chinese billions in Sri Lanka fund battle against Tamil Tigers](#) Times Online
[Huge Humanitarian Operation Under Way in Sri Lanka](#) Voice of America
[BBC News](#) - [Reuters](#) - [AFP](#) - [Xinhua](#)
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Business » [edit](#)

[Buffett Calls Investment Candidates' 2008 Performance Subpar](#)
Bloomberg - 2 hours ago
By Hugh Son, Erik Holm and Andrew Frye May 2 (Bloomberg) -- Billionaire Warren Buffett said all of the candidates to replace him as chief investment officer of Berkshire Hathaway Inc. failed to beat the 38 percent decline of the Standard & Poor's 500 ...
[Buffett offers bleak outlook for US newspapers](#) Reuters
[Buffett Limit CEO pay through embarrassment](#) MarketWatch
[CNBC](#) - [The Associated Press](#) - [guardian.co.uk](#)
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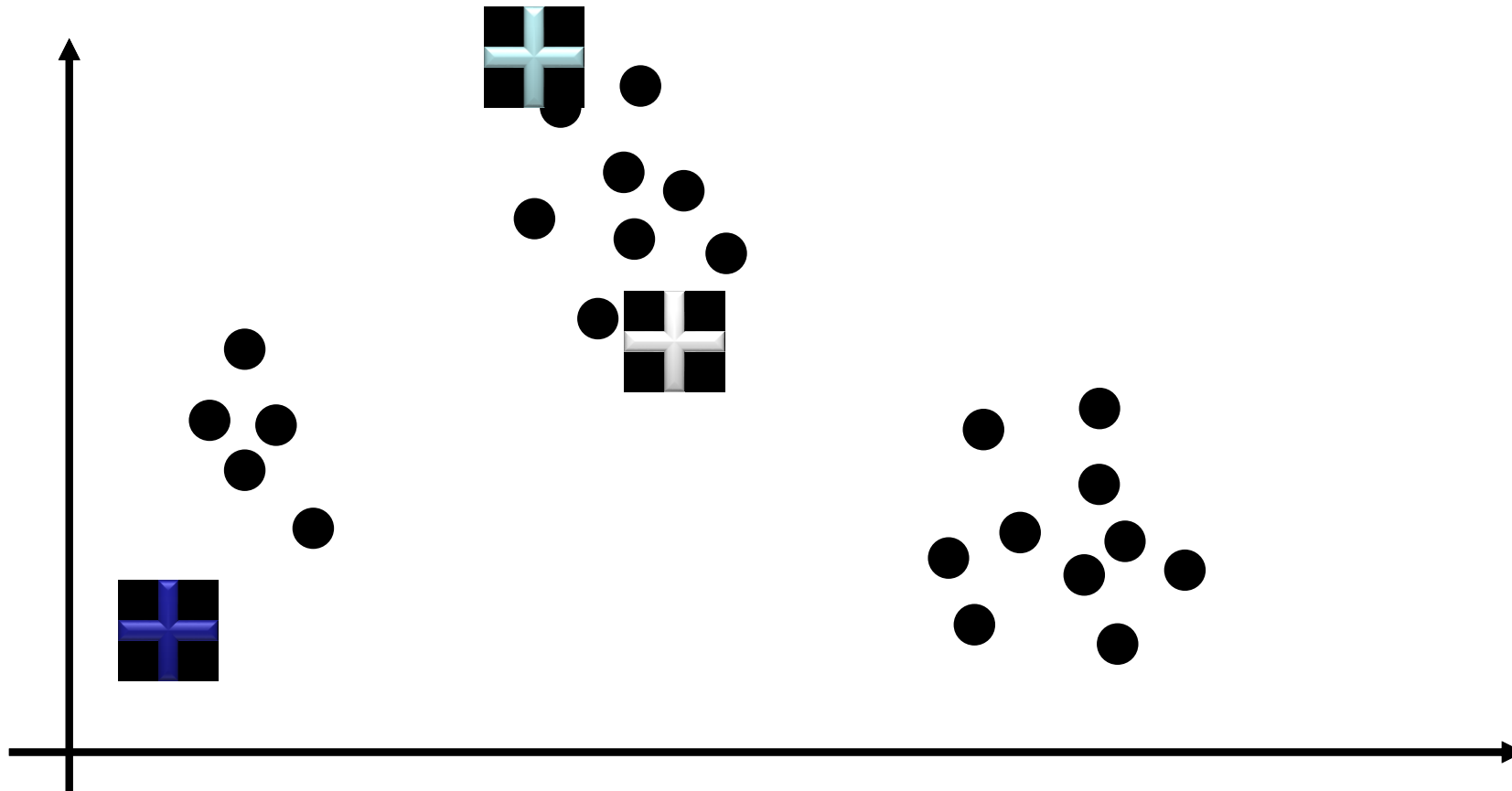
[Chrysler's Fall May Help Administration Reshape GM](#)
New York Times - 5 hours ago
Auto task force members, from left: Treasury's Ron Bloom and Gene Sperling, Labor's Edward Montgomery, and Steve Rattner. BY DAVID E. SANGER and BILL VLASIC
WASHINGTON - Fresh from pushing Chrysler into bankruptcy, President Obama and his economic team ...
[Comment by Gary Chaison](#) Prof. of Industrial Relations, Clark University
[Bankruptcy reality sets in for Chrysler, workers](#) Detroit Free Press
[Washington Post](#) - [Bloomberg](#) - [CNNMoney.com](#)
[all 11,028 news articles »](#)

K-Means



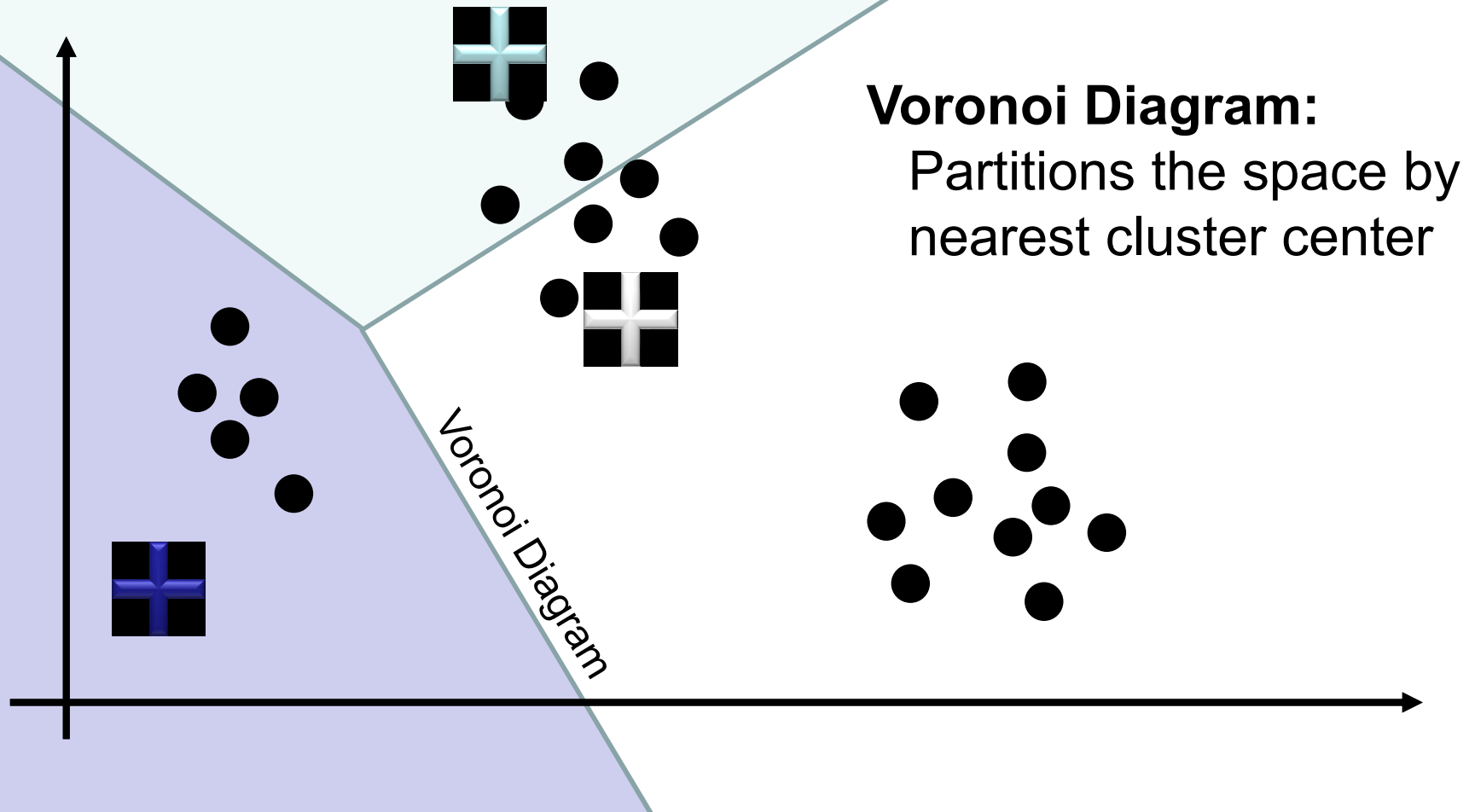
K-Means Clustering: *Intuition*

- Input K: The number of clusters to find
- Pick an initial set of points as cluster centers



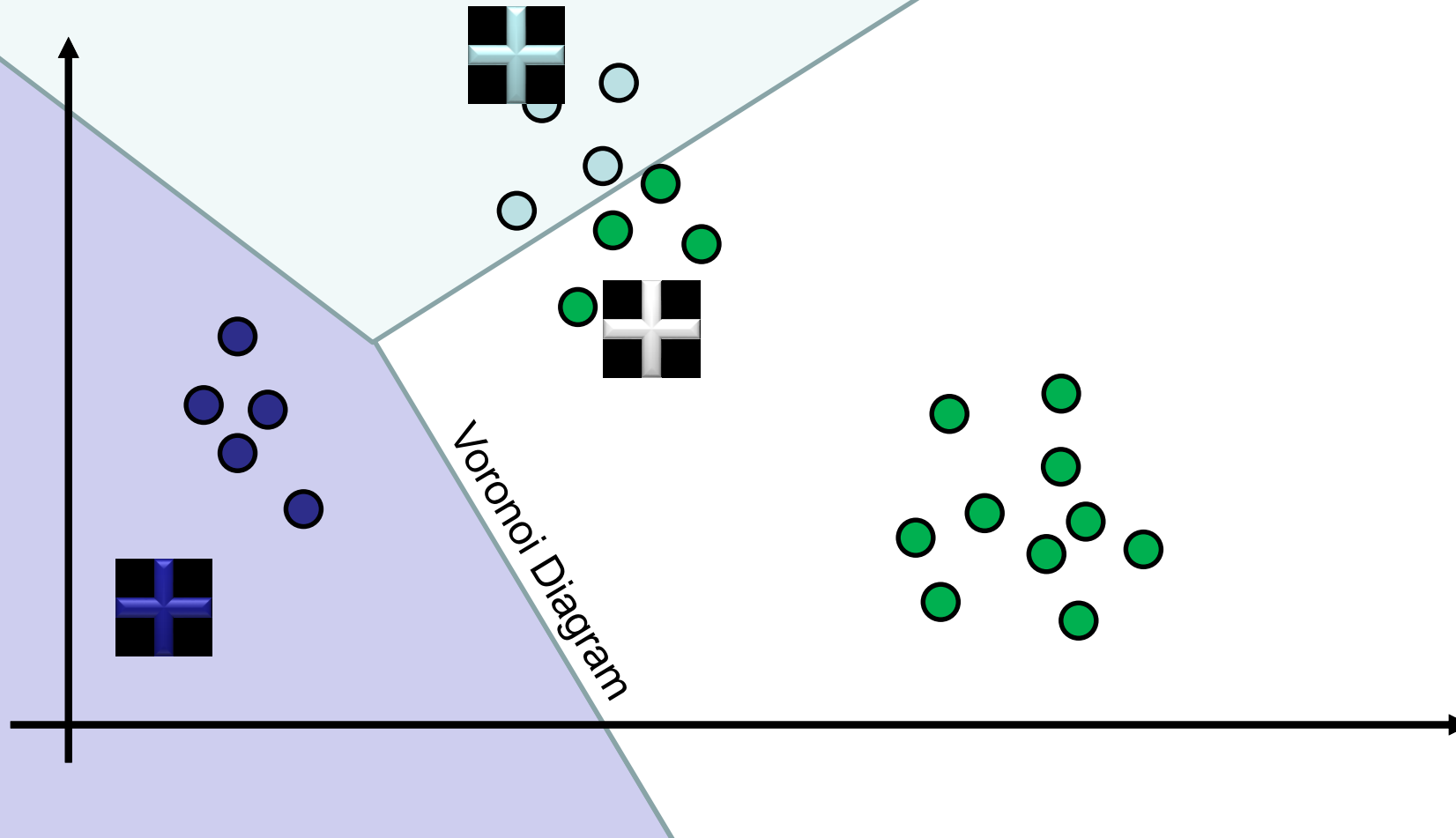
K-Means Clustering: *Intuition*

- For each data point find the cluster nearest center



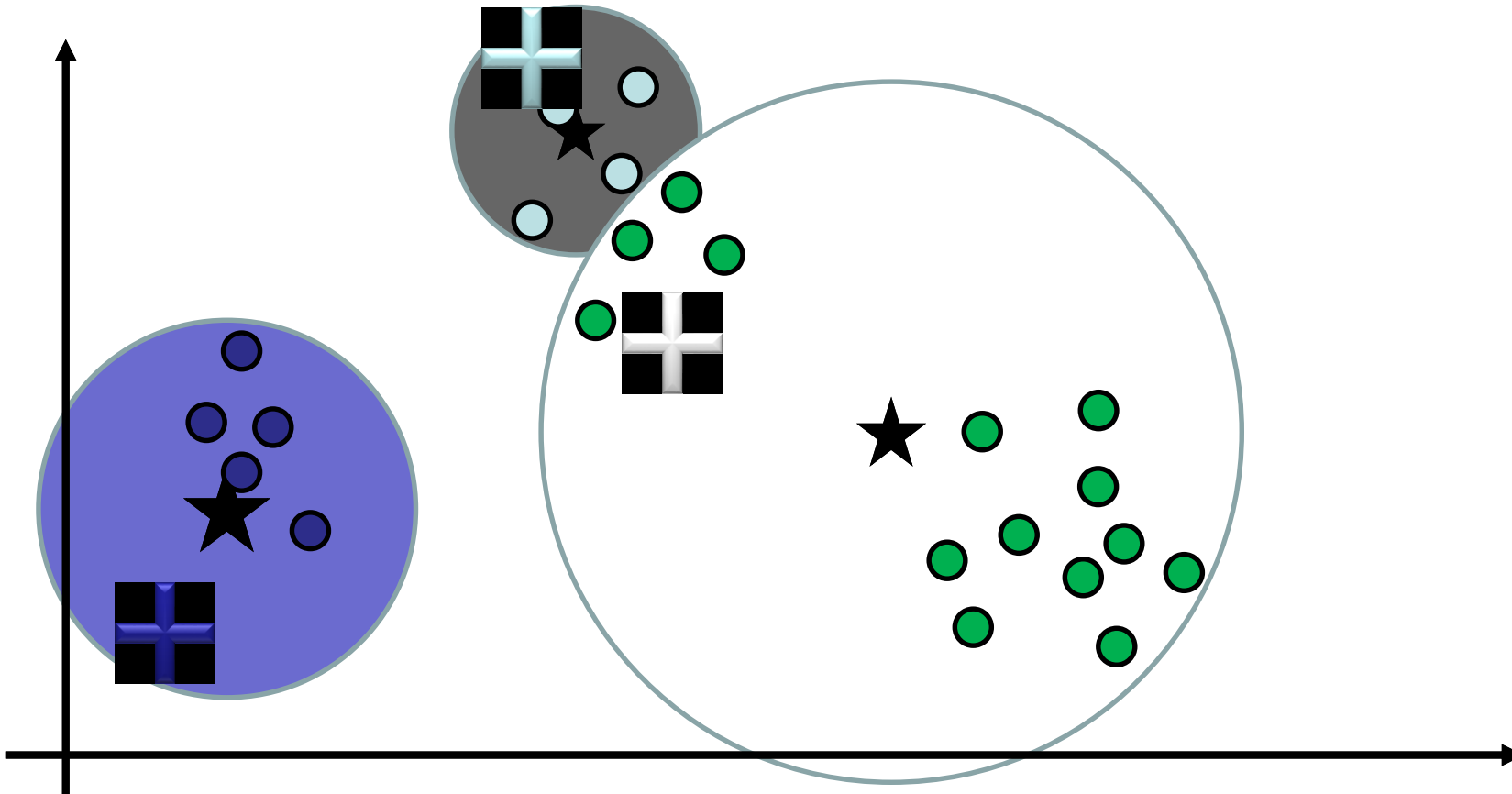
K-Means Clustering: *Intuition*

- For each data point find the cluster nearest center



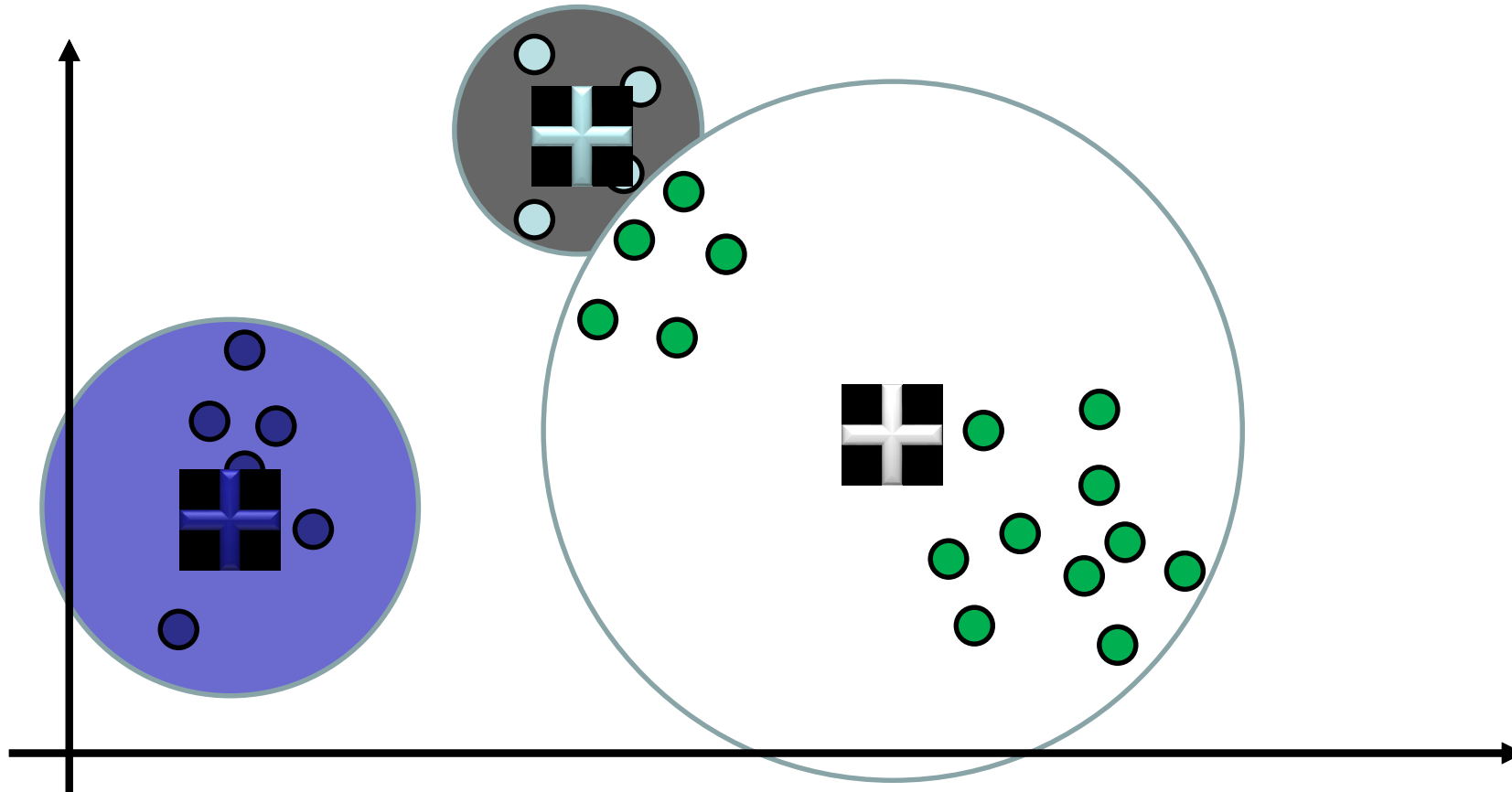
K-Means Clustering: *Intuition*

- Compute mean of points in each “cluster”



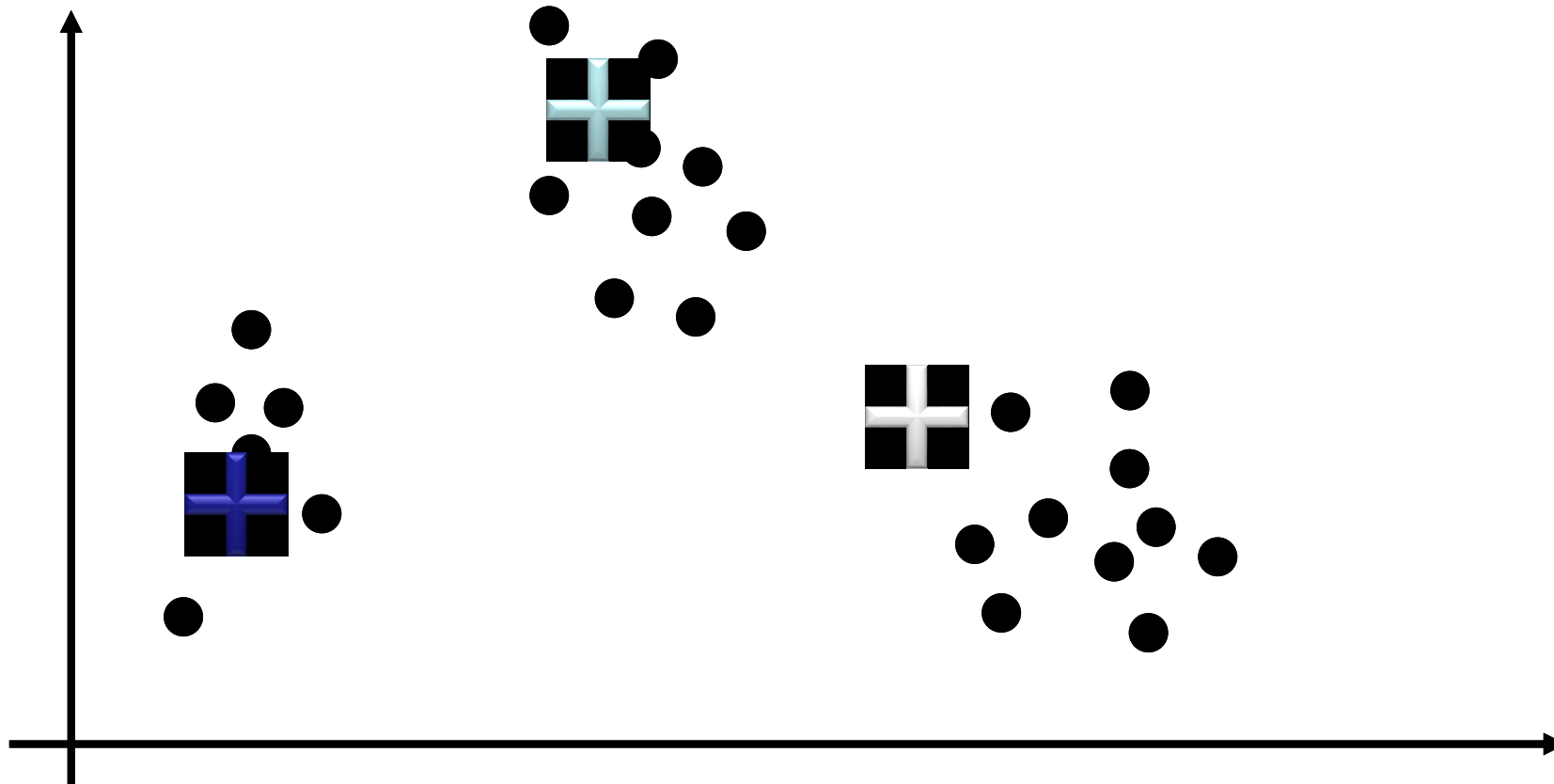
K-Means Clustering: *Intuition*

- Adjust cluster centers to be the mean of the cluster



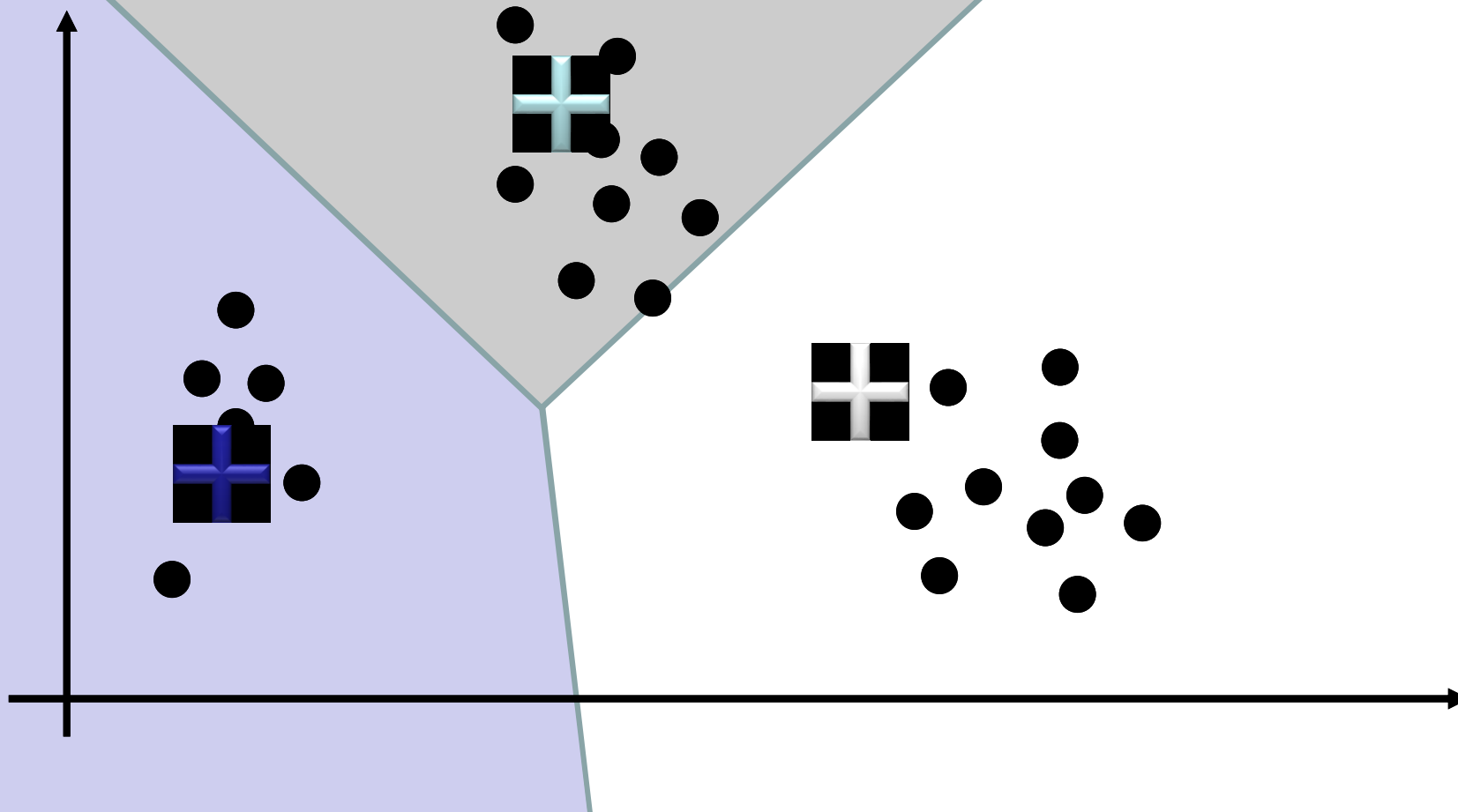
K-Means Clustering: *Intuition*

- Improved?
- Repeat



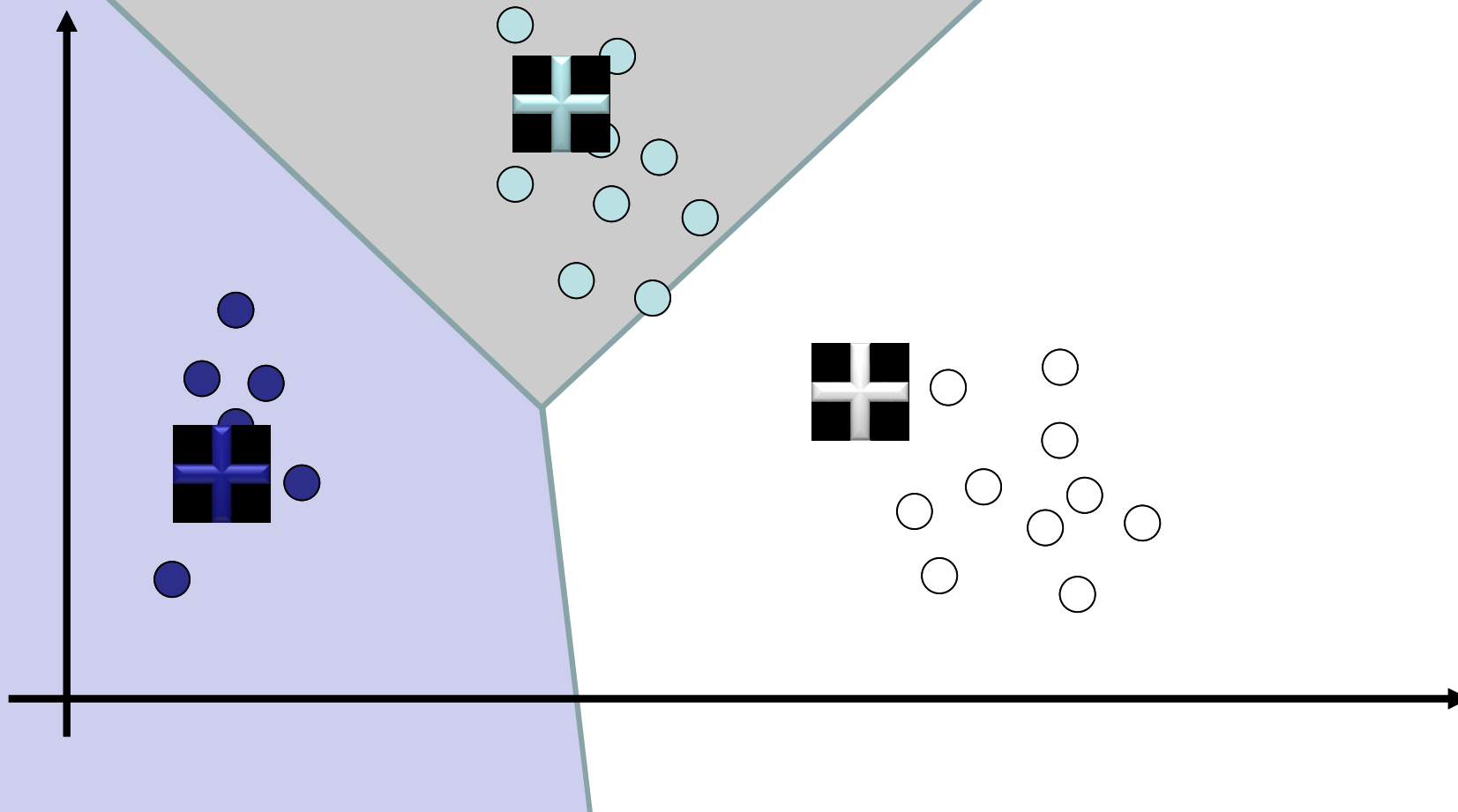
K-Means Clustering: *Intuition*

- Assign Points



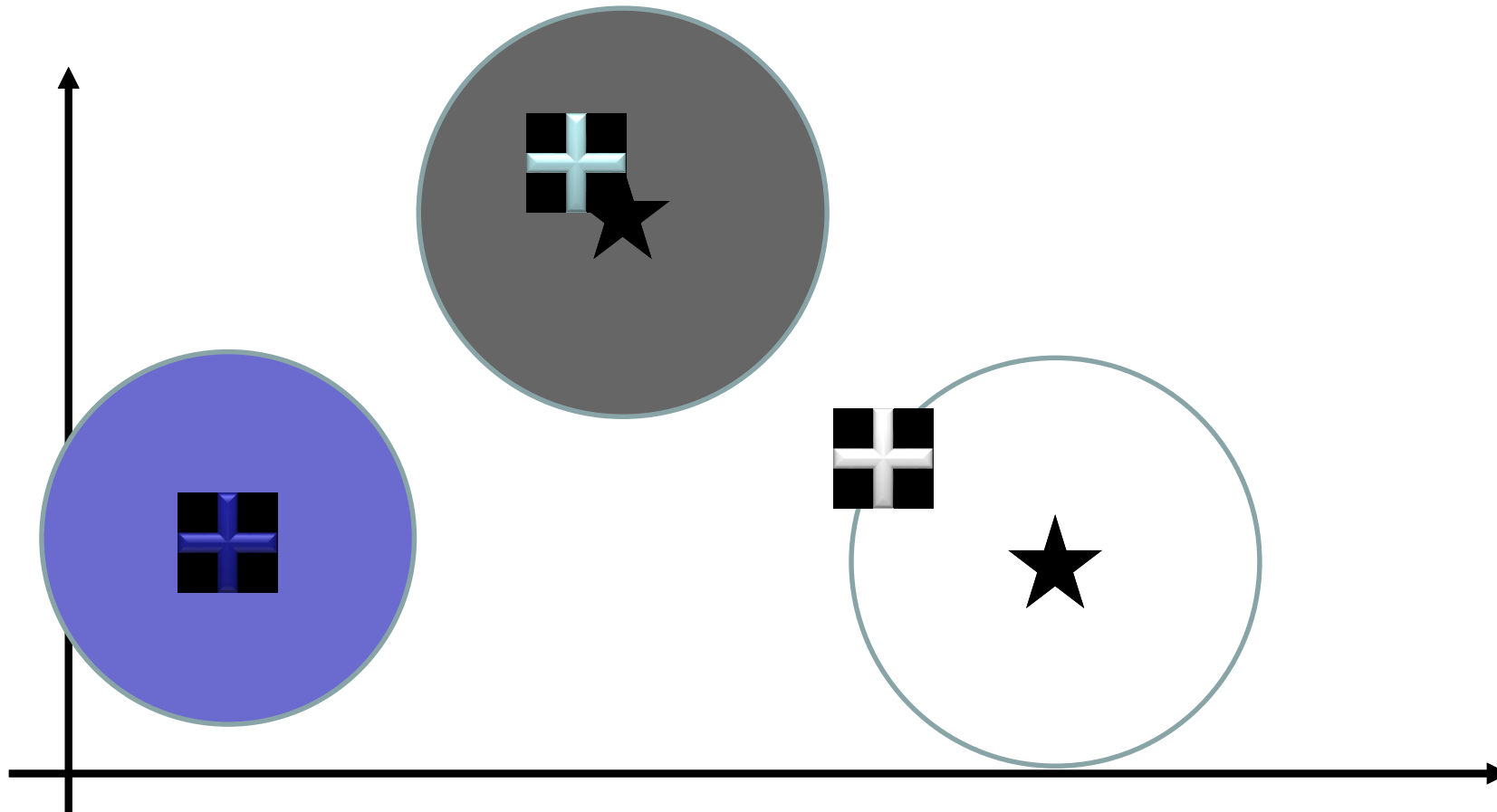
K-Means Clustering: *Intuition*

- Assign Points



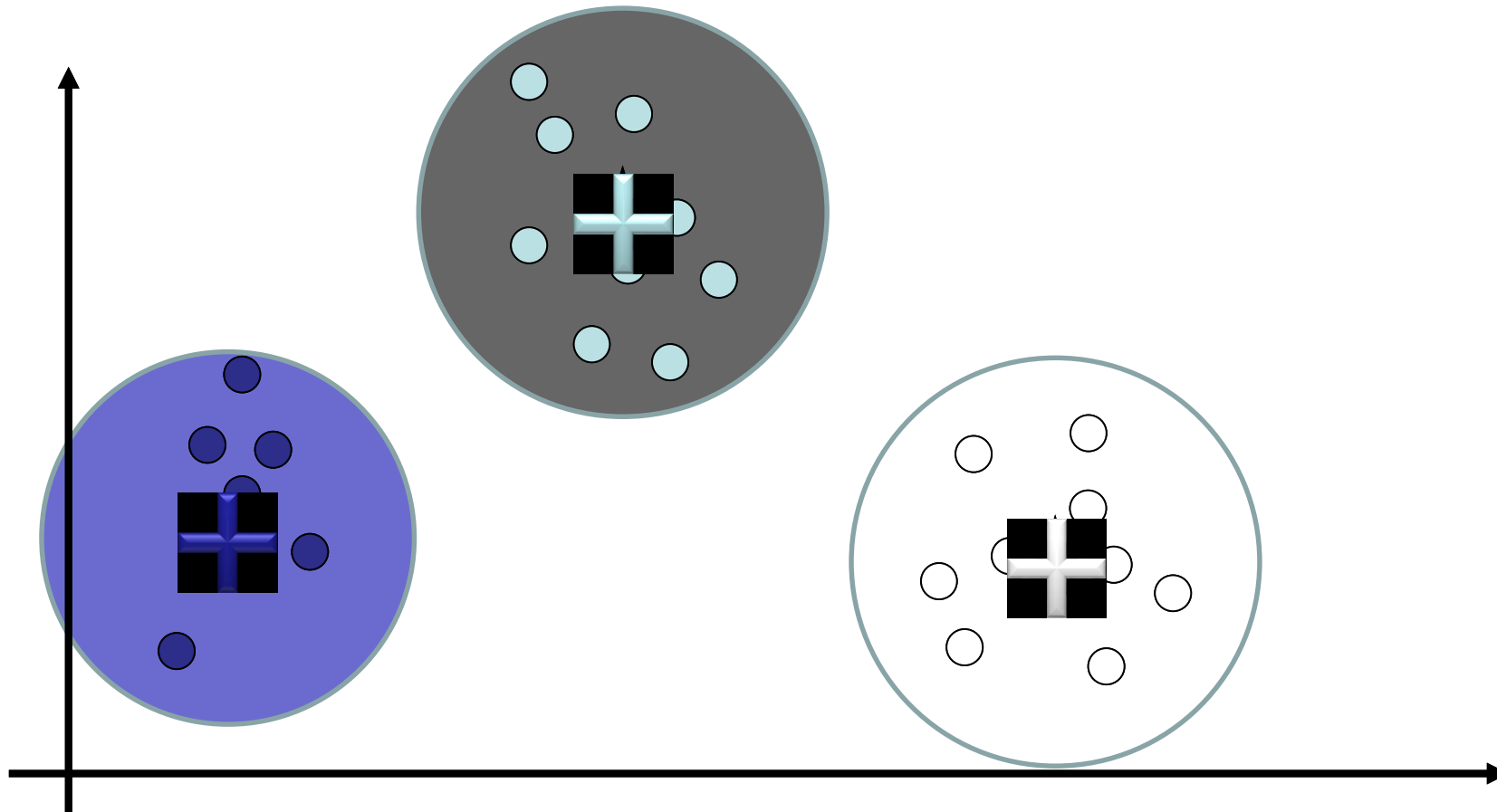
K-Means Clustering: *Intuition*

- Compute cluster means



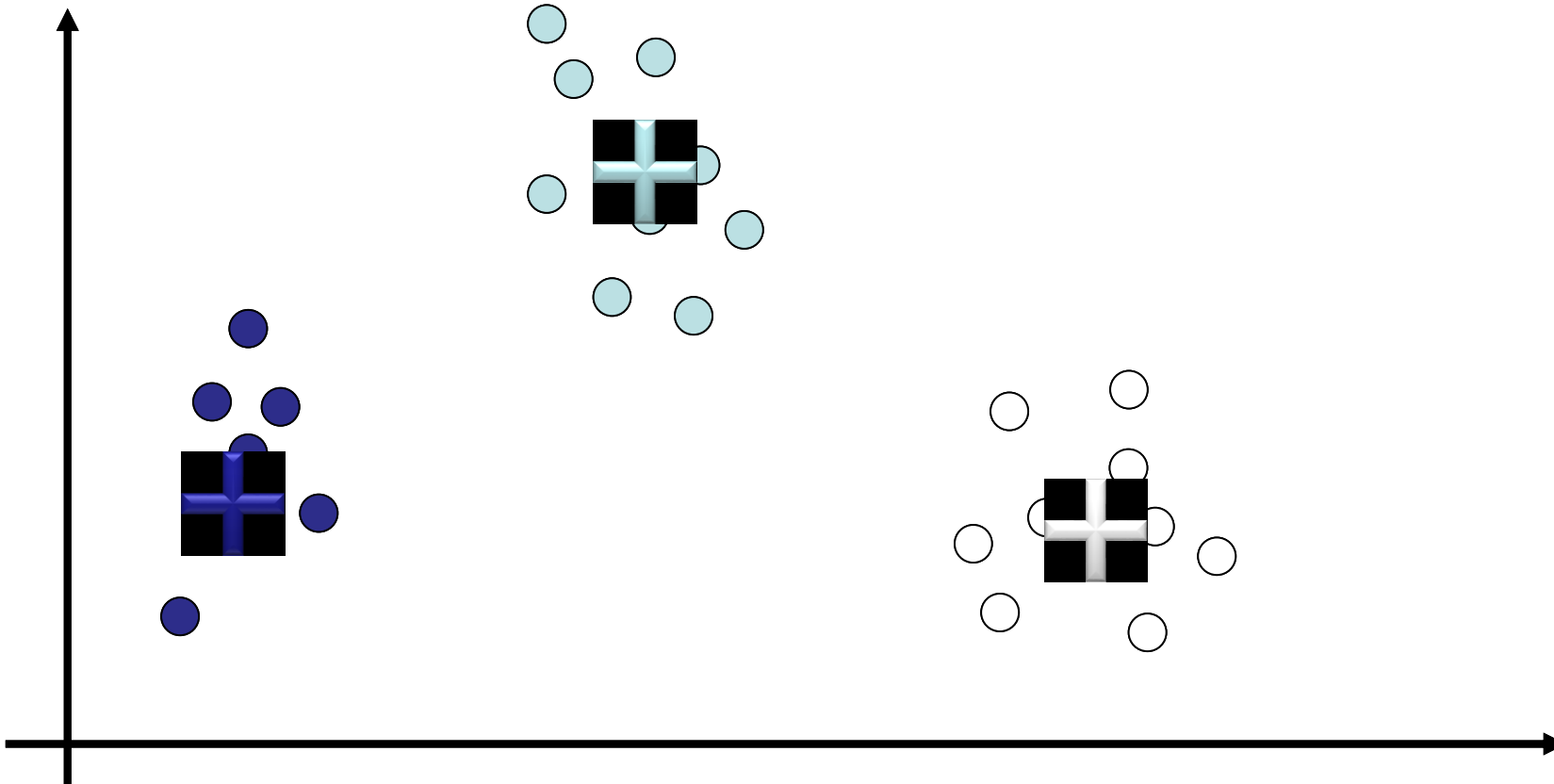
K-Means Clustering: *Intuition*

- Update cluster centers



K-Means Clustering: *Intuition*

- Repeat?
 - Yes to check that nothing changes → Converged!

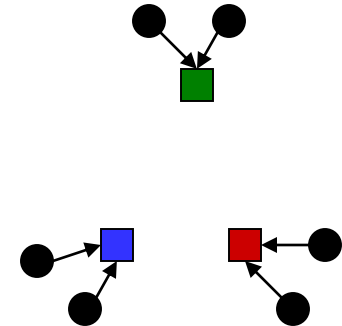


K-Means as Optimization

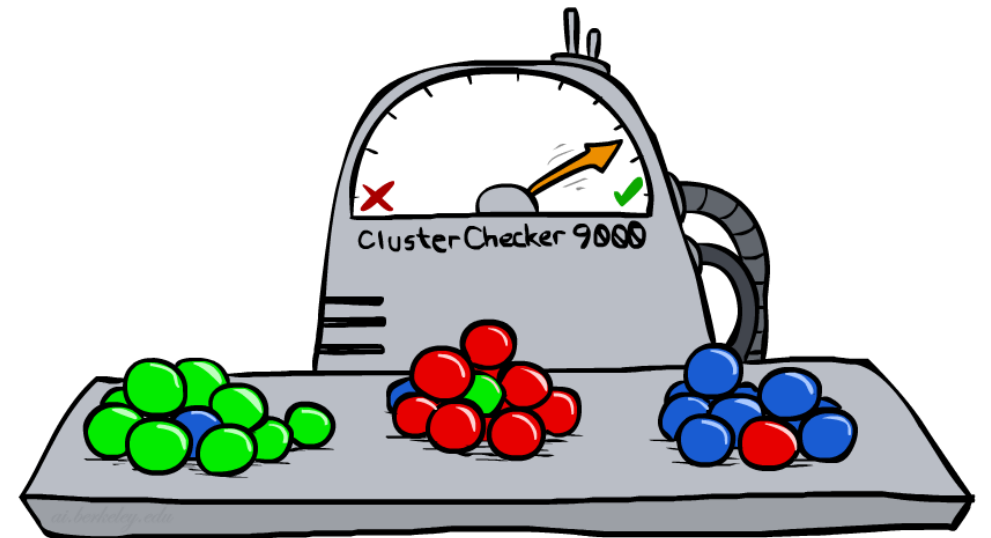
- Consider the total distance to the means:

$$\phi(\{x_i\}, \{a_i\}, \{c_k\}) = \sum_i \text{dist}(x_i, c_{a_i})$$

points assignments means squared Euclidean distance



- Two stages each iteration:
 - Update assignments: fix means c , change assignments a
 - Update means: fix assignments a , change means c
- Each step cannot increase ϕ



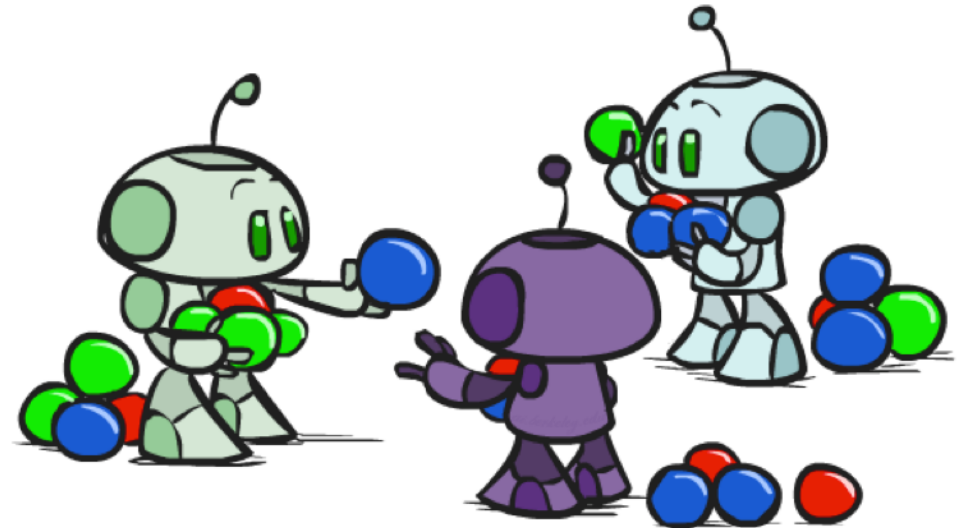
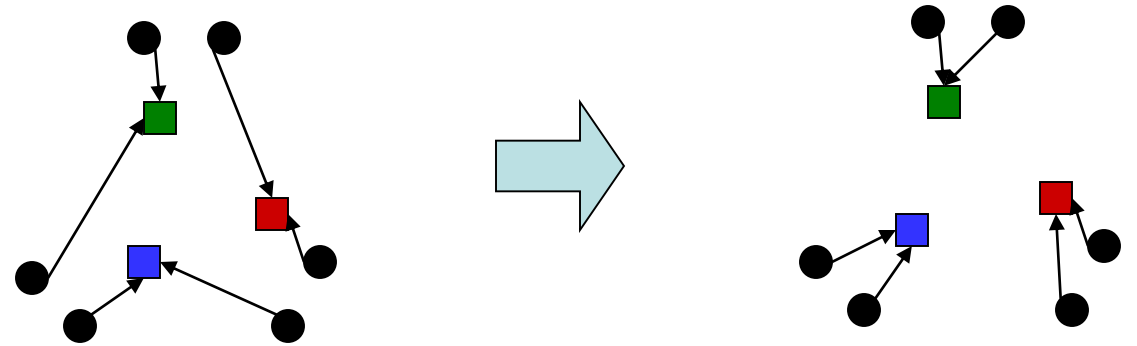
Phase I: Update Assignments

- For each point, re-assign to closest mean:

$$a_i = \operatorname{argmin}_k \text{dist}(x_i, c_k)$$

- Cannot increase total distance phi!

$$\phi(\{x_i\}, \{a_i\}, \{c_k\}) = \sum_i \text{dist}(x_i, c_{a_i})$$

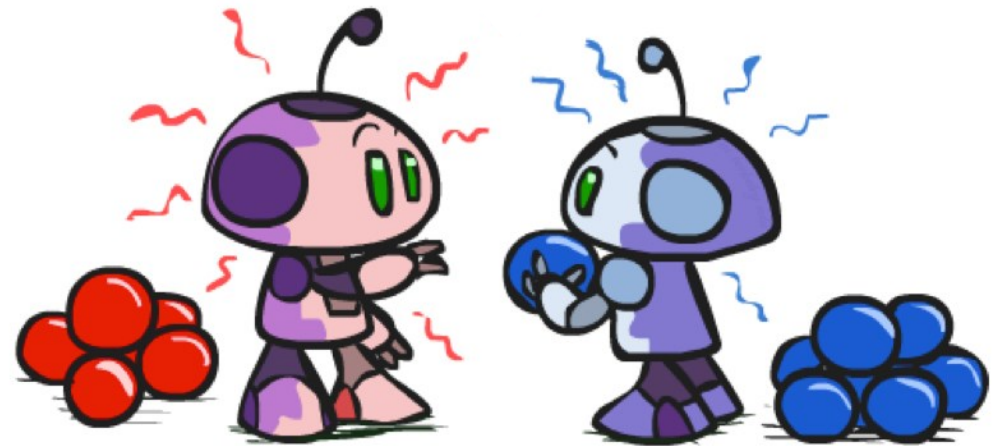
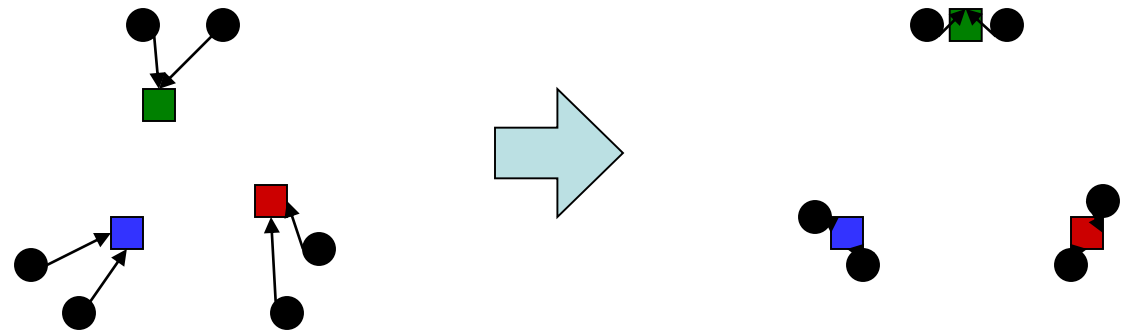


Phase II: Update Means

- Move each mean to the average of its assigned points:

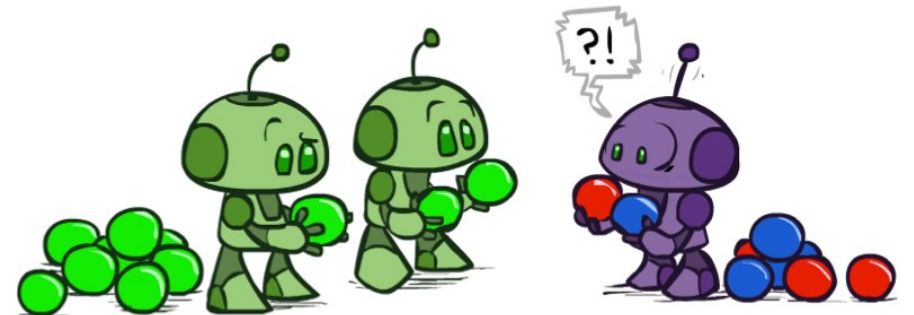
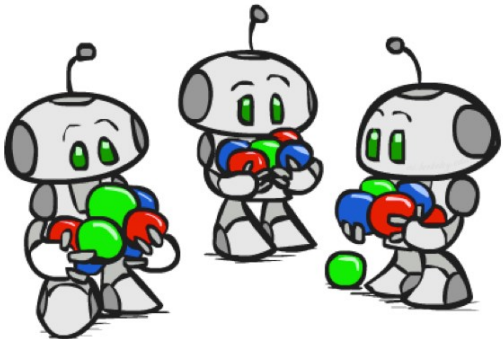
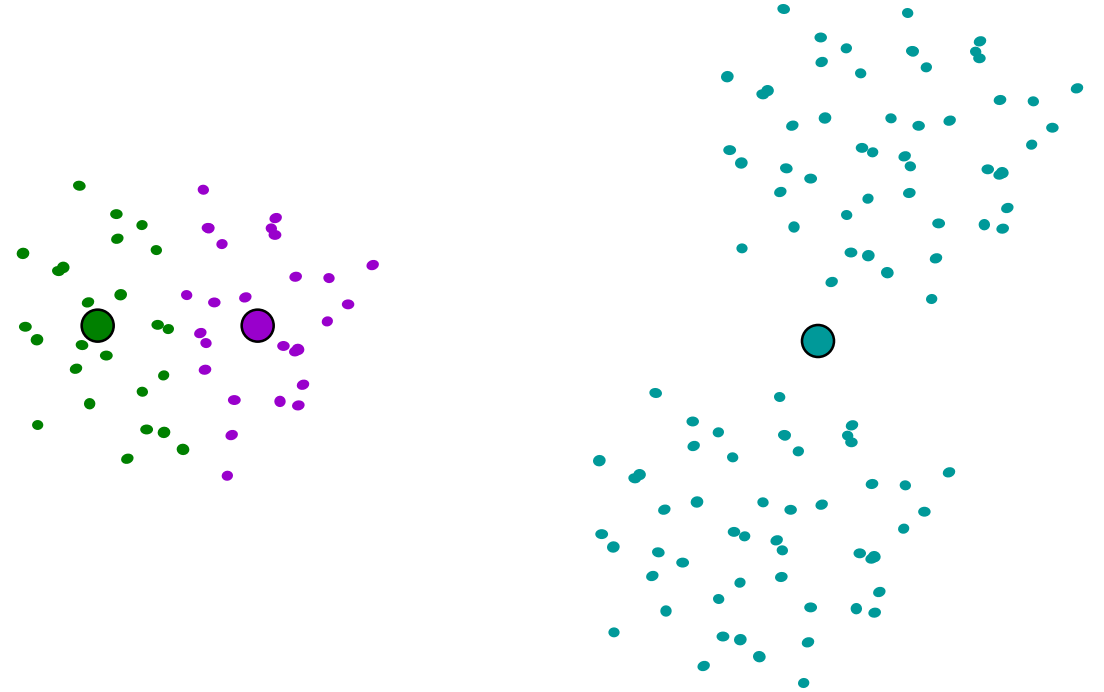
$$c_k = \frac{1}{|\{i : a_i = k\}|} \sum_{i:a_i=k} x_i$$

- Also cannot increase total distance
 - Fun fact: the point y with minimum squared Euclidean distance to a set of points $\{x\}$ is their mean

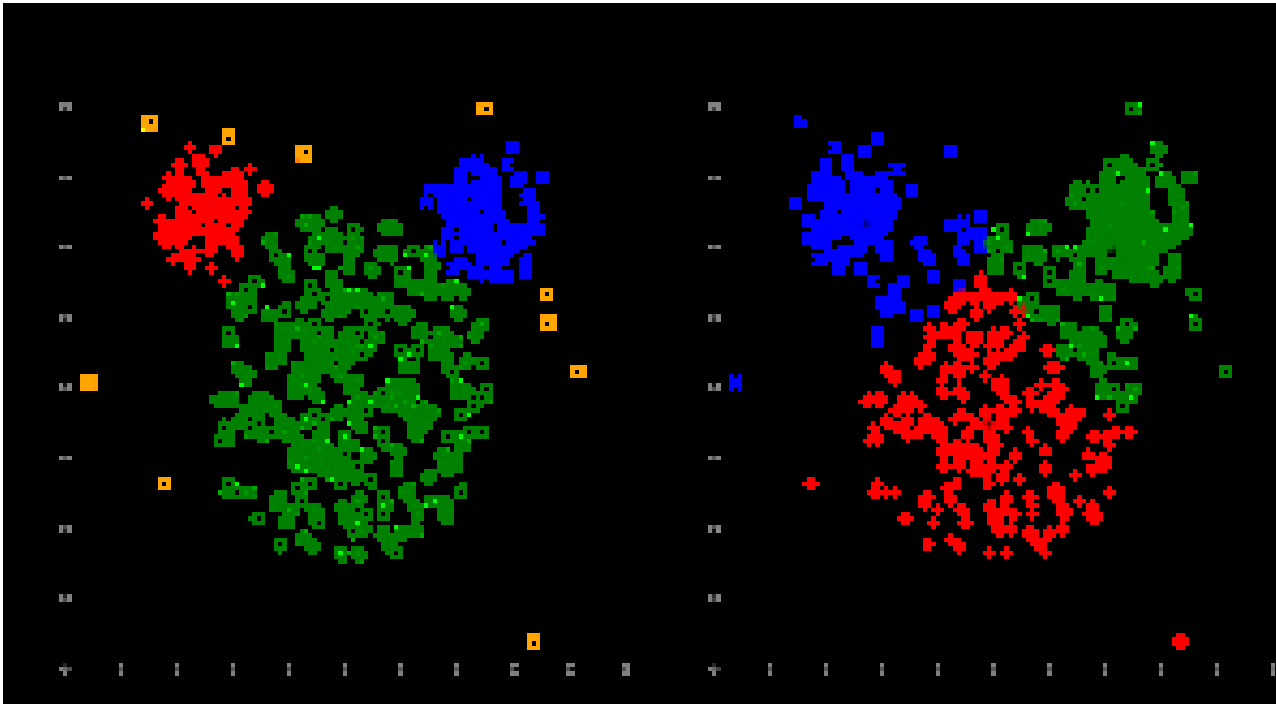


Initialization

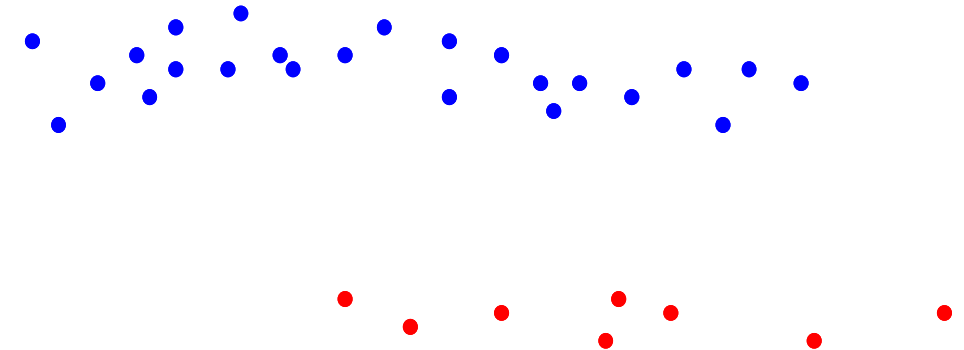
- K-means is non-deterministic
 - Requires initial means
 - It does matter what you pick!
 - What can go wrong?
 - Local optima



Inductive Bias



Equally Sized Clusters

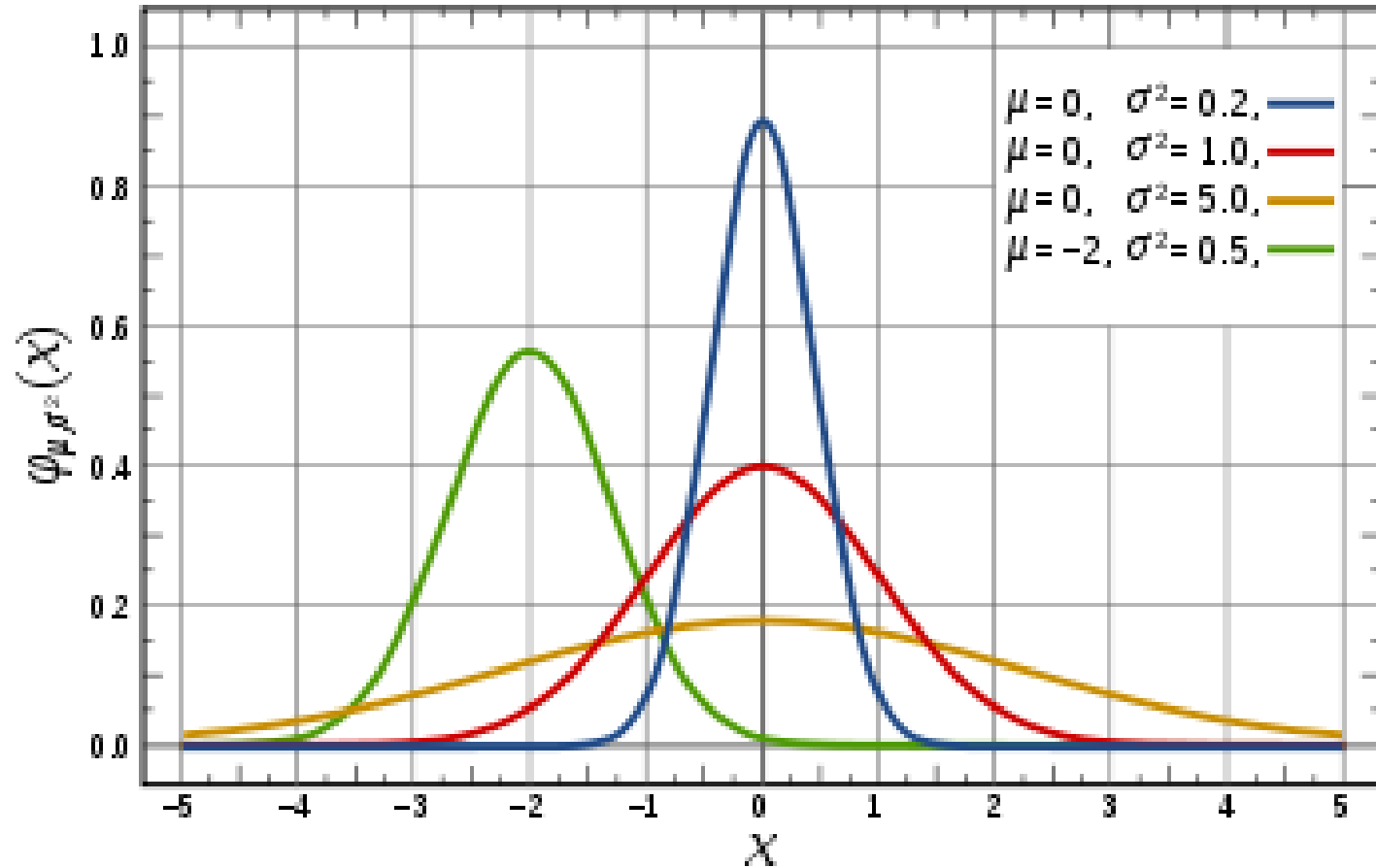


Circular Clusters

Probabilistic Clustering

- Try a probabilistic model!
 - allows overlaps, clusters of different sizes/shapes, etc.
- Gaussian mixture model (GMM)
 - also called Mixture of Gaussians

Review: Gaussians



$$P(x \mid \mu, \sigma) = \frac{1}{\sigma \sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

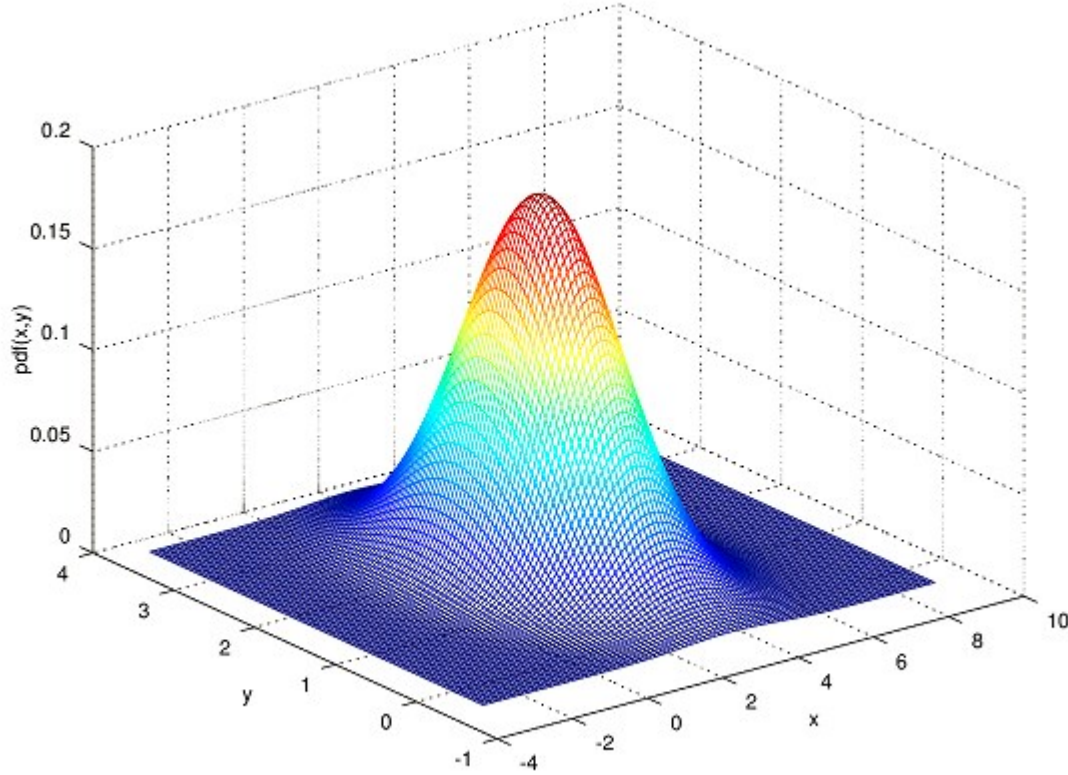
Learning Gaussian Parameters

Given fully-observable data:

$$\hat{\mu}_{MLE} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2$$

Multivariate Gaussians



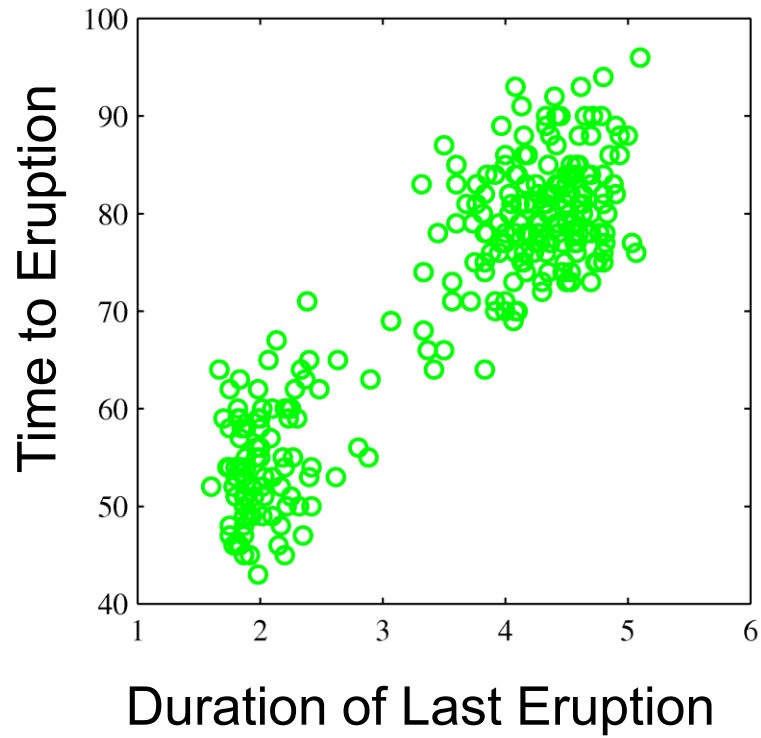
Covariance matrix Σ :
degree to which x_i vary
together

$$P(X = \mathbf{x}) = \frac{1}{(2\pi)^{m/2} \|\Sigma\|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right]$$

Two purple arrows originate from the text above. One arrow points to the constant term $\frac{1}{(2\pi)^{m/2} \|\Sigma\|^{1/2}}$, and the other arrow points to the covariance matrix Σ in the exponent.

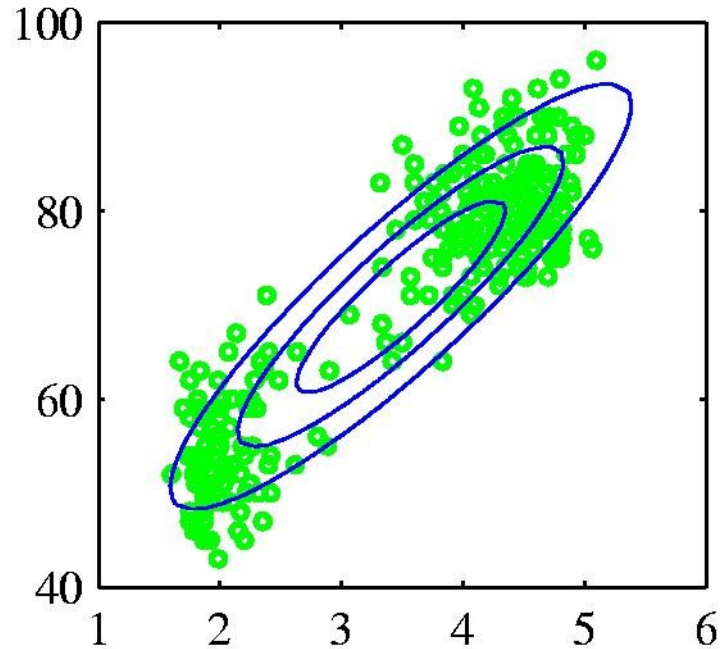
Mixtures of Gaussians

- Old Faithful Data Set

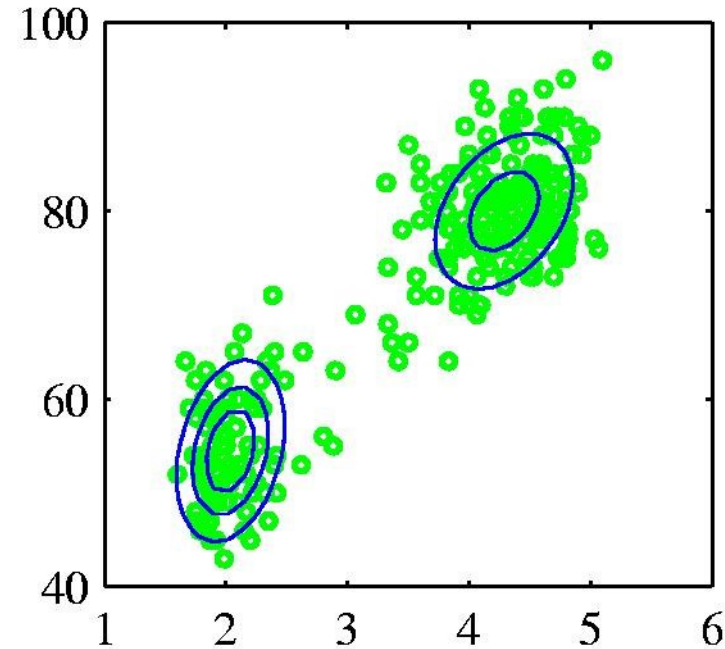


Mixtures of Gaussians

- Old Faithful Data Set



Single Gaussian



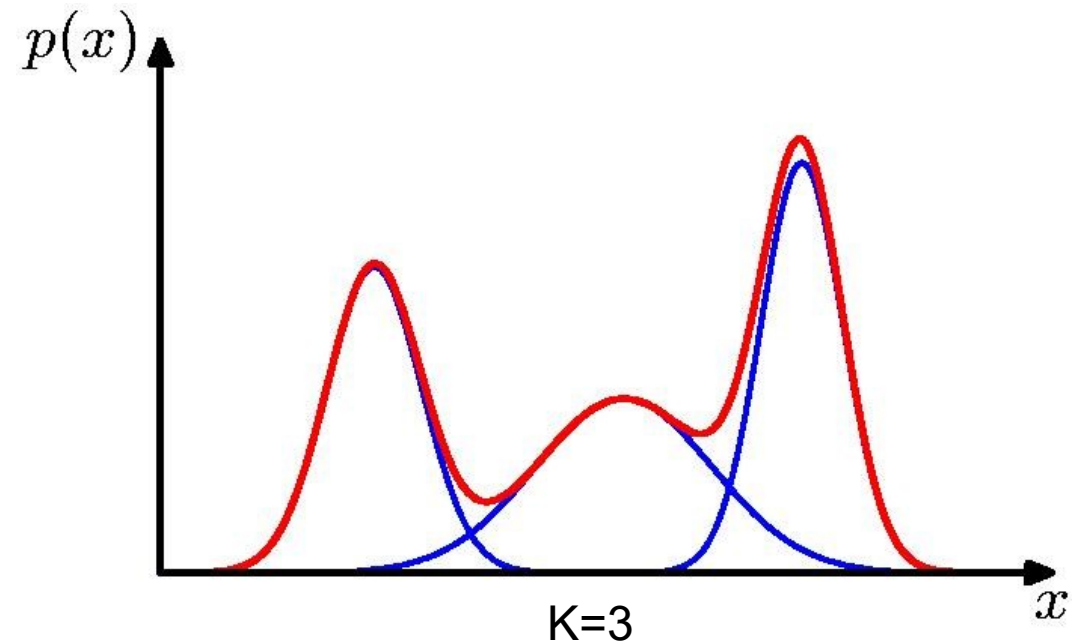
Mixture of two
Gaussians

Mixtures of Gaussians

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \underbrace{\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{Component}}$$

Mixing coefficient

$$\forall k : \pi_k \geq 0 \quad \sum_{k=1}^K \pi_k = 1$$

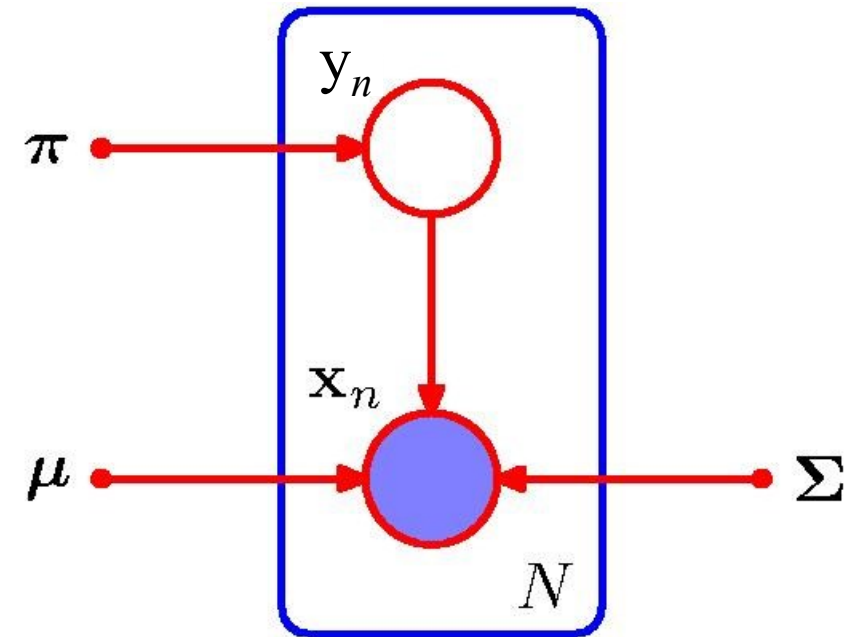


Gaussian mixture model

- $P(Y)$: Distribution over k components (clusters)
- $P(X|Y)$: Each component generates data from a **multivariate Gaussian** with mean μ_i and covariance matrix Σ_i

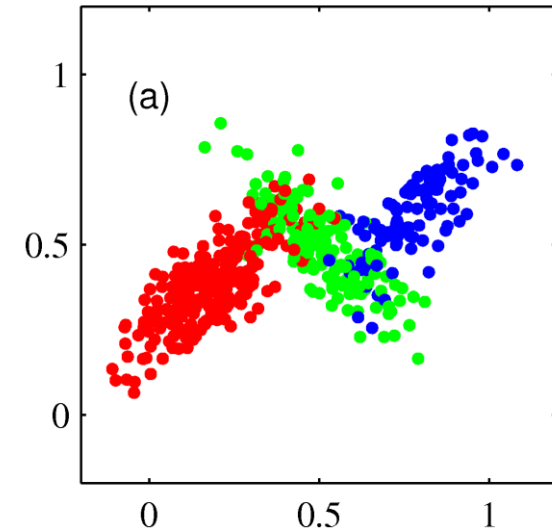
Each data point is sampled from a
generative process:

1. Choose component i with probability π_i
2. Generate data point from $N(\mathbf{x} | \mu_i, \Sigma_i)$



Supervised learning for GMM

- We observe both the data points and their labels (generated from which Gaussian components)
- How do we estimate parameters of GMM?



Supervised learning for GMM

- We observe both the data points and their labels (generated from which Gaussian components)
- How do we estimate parameters of GMM?
- Objective: maximize the likelihood

$$\prod_j P(y_j = i, \mathbf{x}_j) = \prod_j \pi_i N(\mathbf{x}_j | \mu_i, \Sigma_i)$$

- Closed form solution:
 - m data points. For component i , suppose we have n data points with label i .

$$\mu_i = \frac{1}{n} \sum_{j=1}^n \mathbf{x}_j$$

$$\Sigma_i = \frac{1}{n} \sum_{j=1}^n (\mathbf{x}_j - \mu_i)(\mathbf{x}_j - \mu_i)^T$$

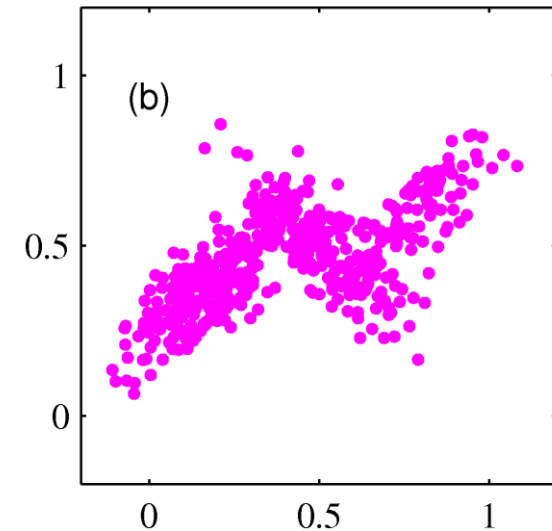
$$\pi_i = \frac{n}{m}$$

Unsupervised learning for GMM

- In clustering, we don't know the labels Y !
- Maximize marginal likelihood:

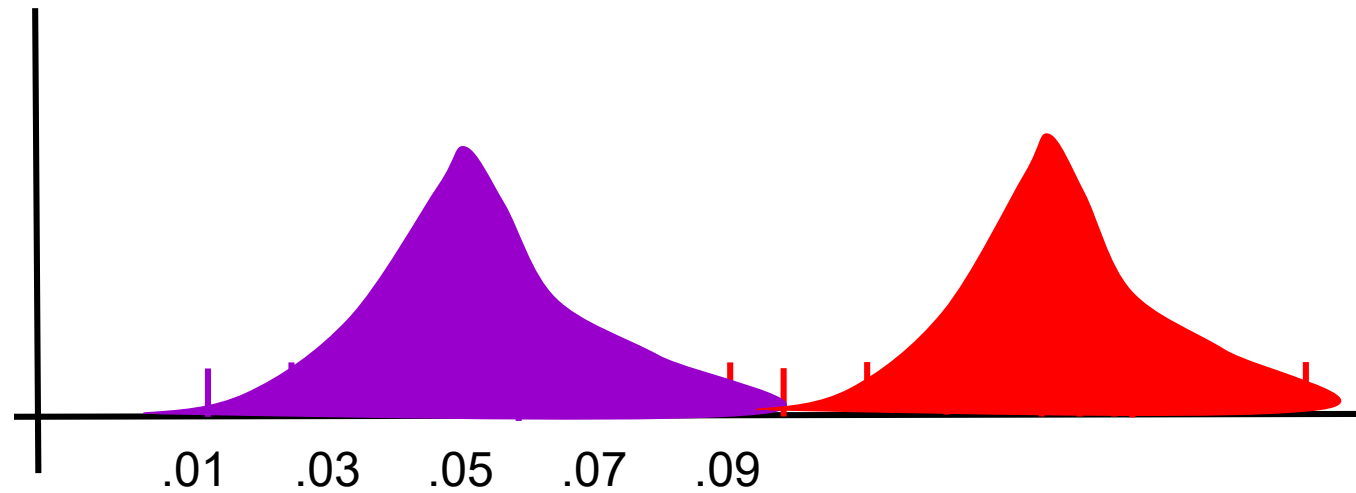
$$\prod_j P(\mathbf{x}_j) = \prod_j \sum_i P(y_j = i, \mathbf{x}_j) = \prod_j \sum_i \pi_i N(\mathbf{x}_j | \mu_i, \Sigma_i)$$

- How do we optimize it?
 - No closed form solution

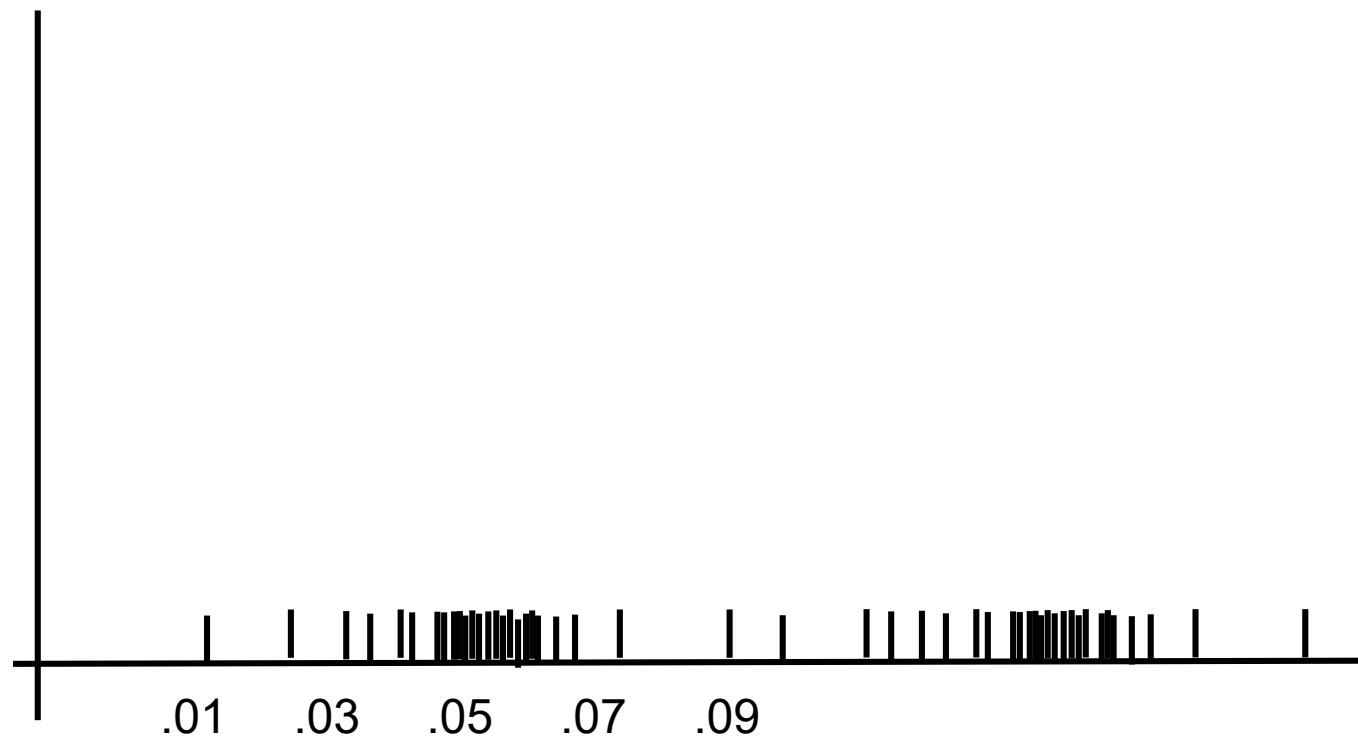


Simplest Example

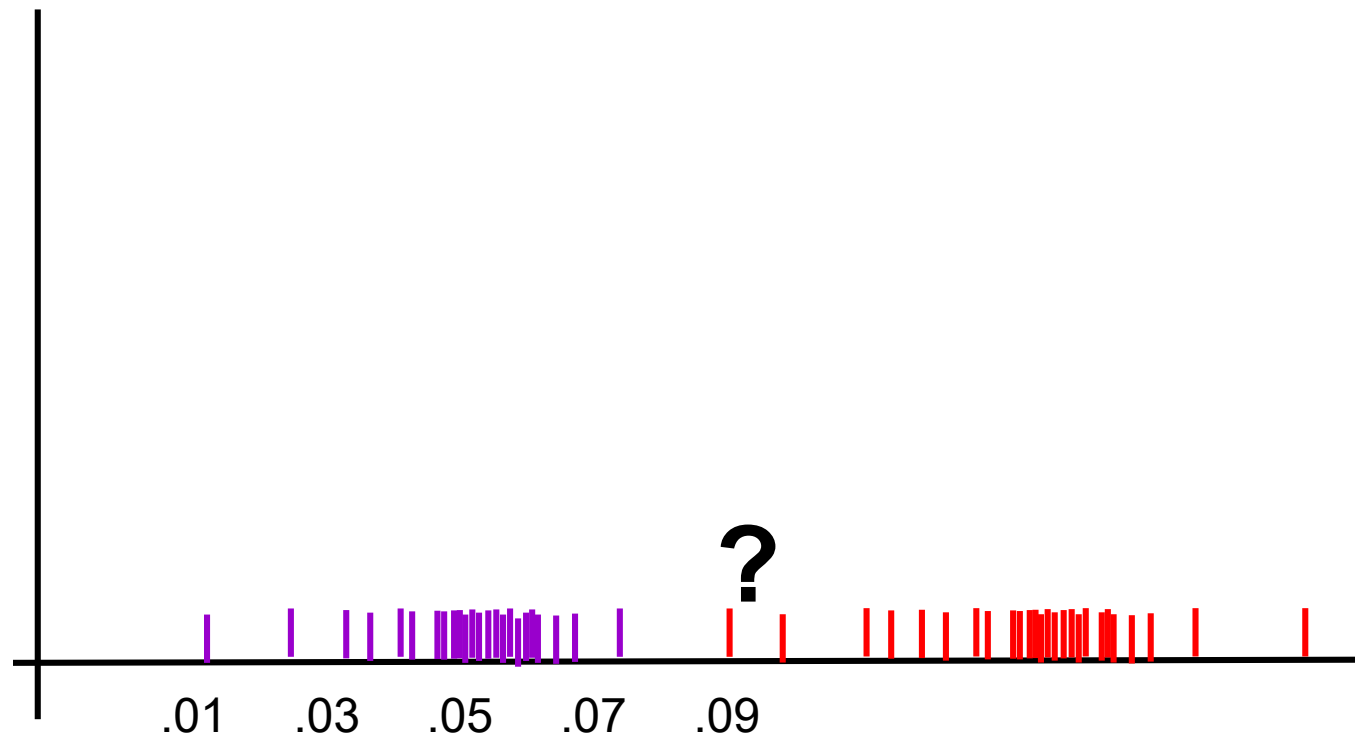
Mixture of two distributions



Input Looks Like

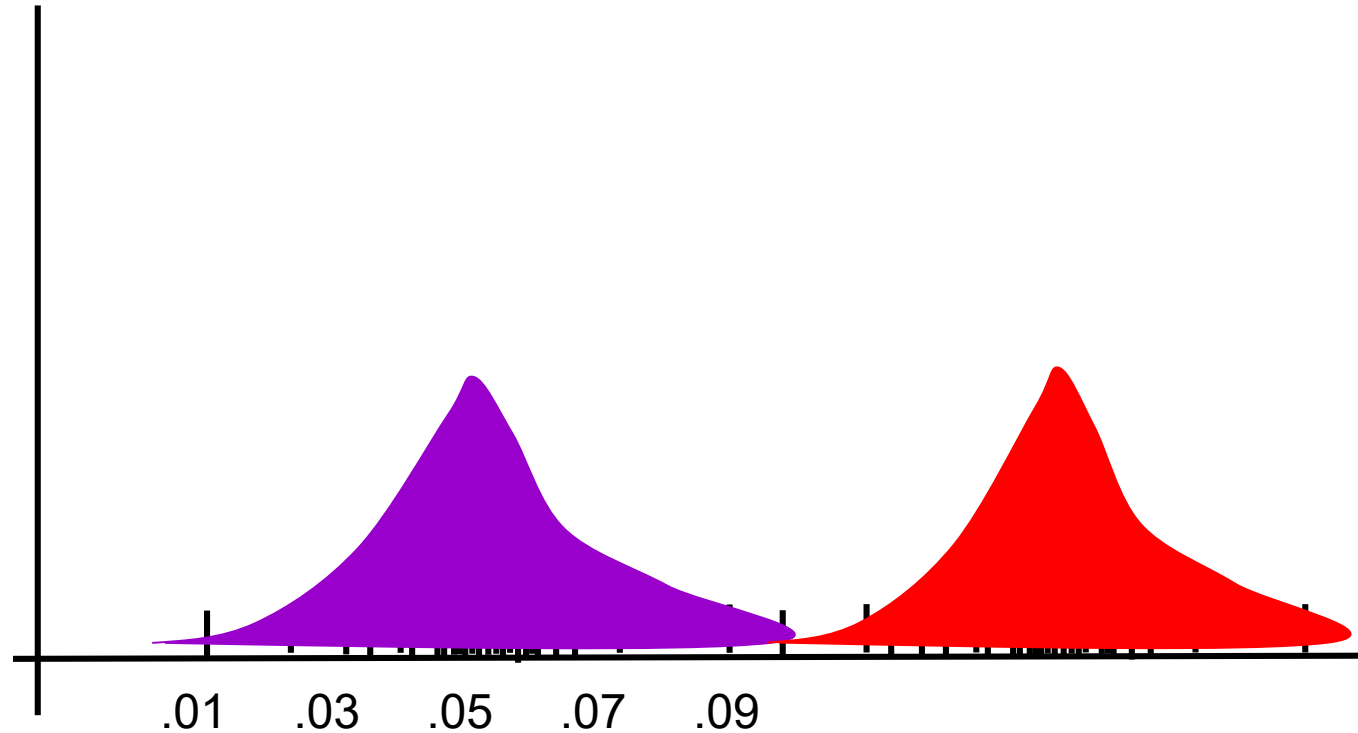


We Want to Predict



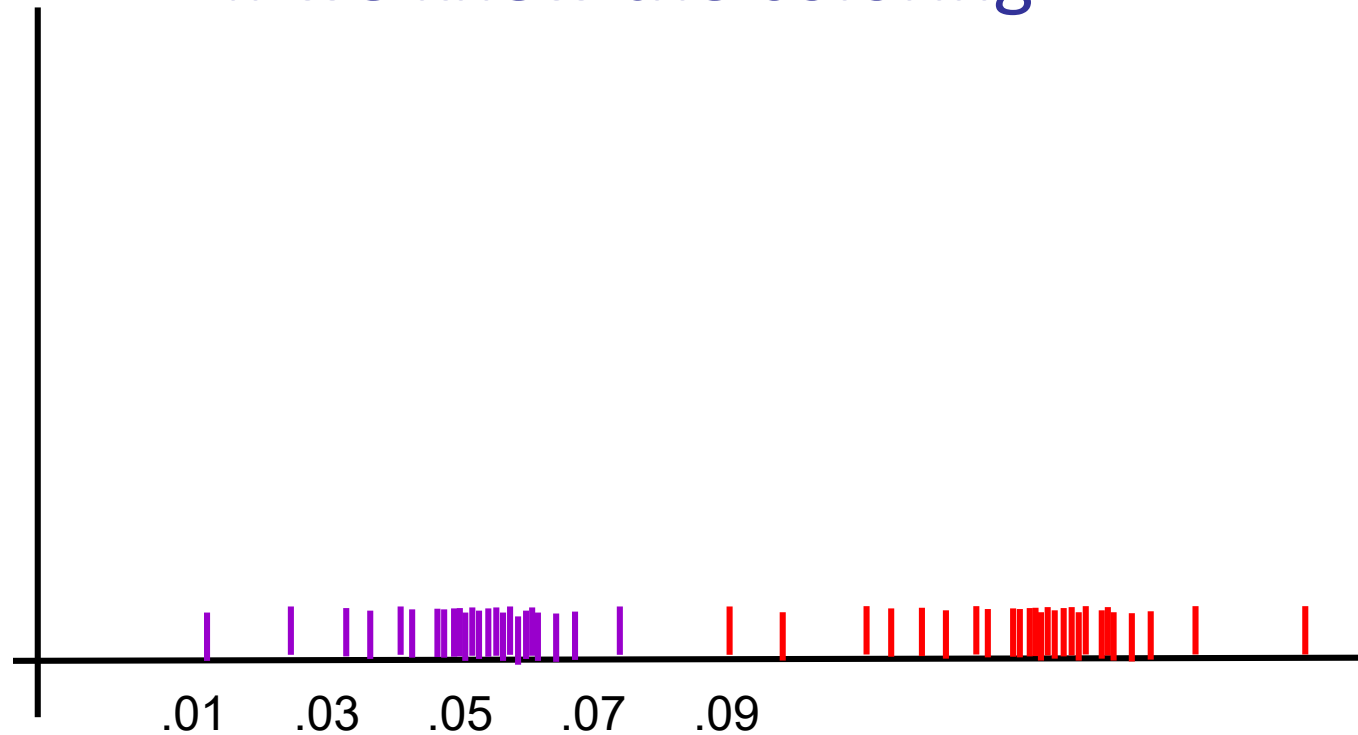
Chicken & Egg

Note that coloring instances would be easy
if we knew Gaussians....



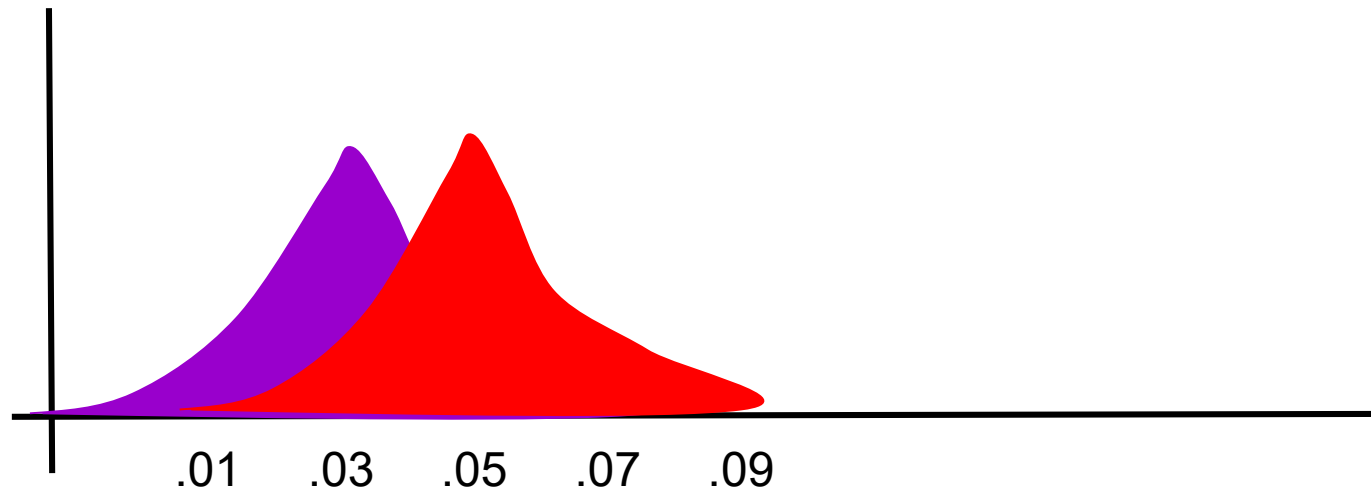
Chicken & Egg

And finding the Gaussians would be easy
If we knew the coloring



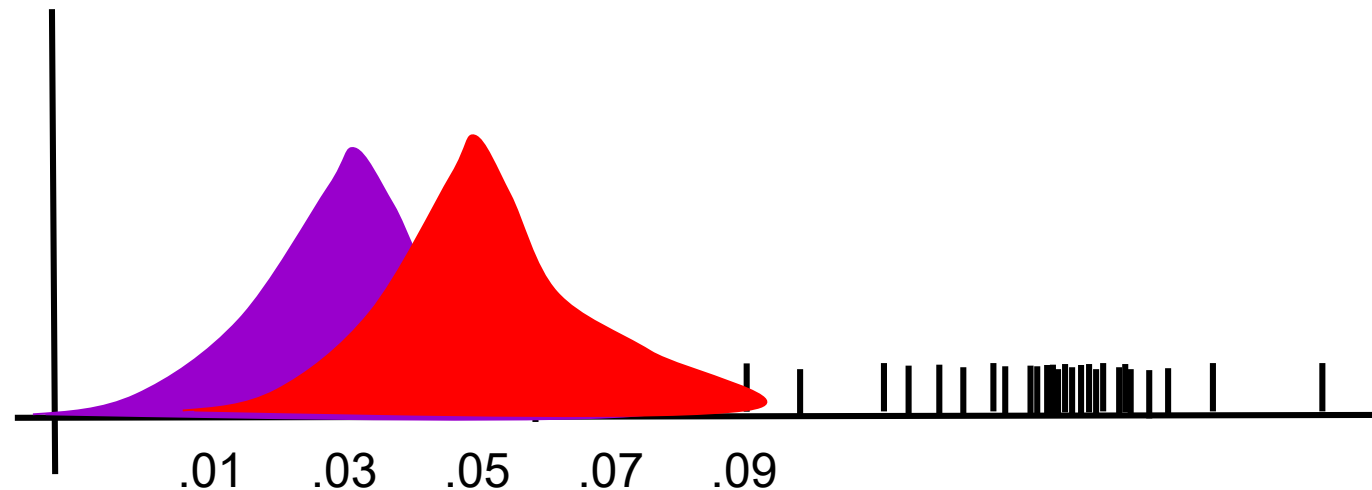
Expectation Maximization (EM)

- Pretend we do know the parameters
 - Initialize randomly



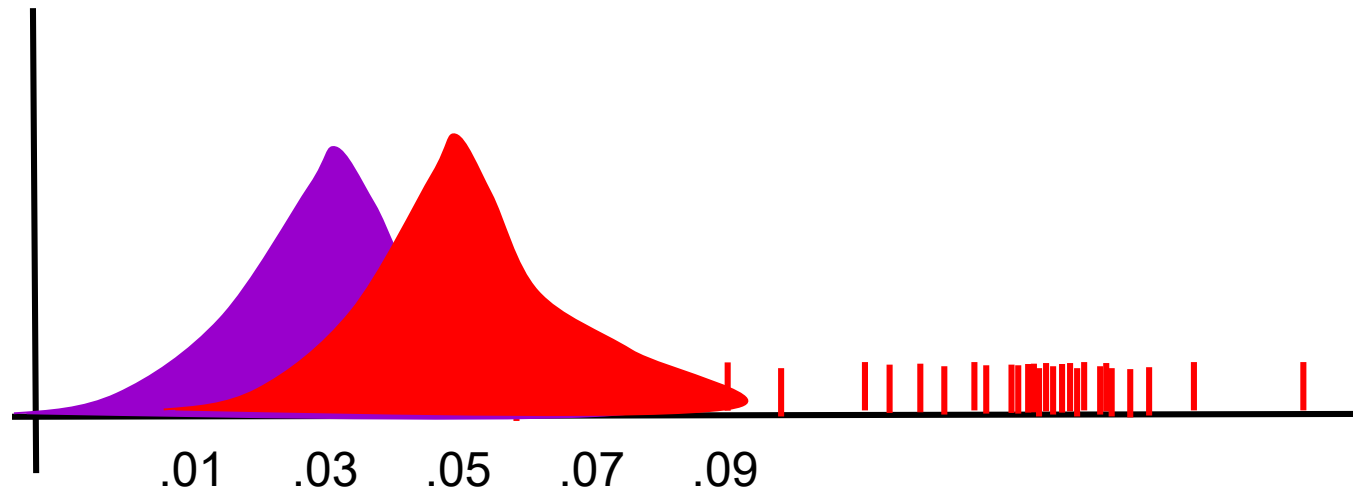
Expectation Maximization (EM)

- [E step] Compute probability of each instance having each possible label



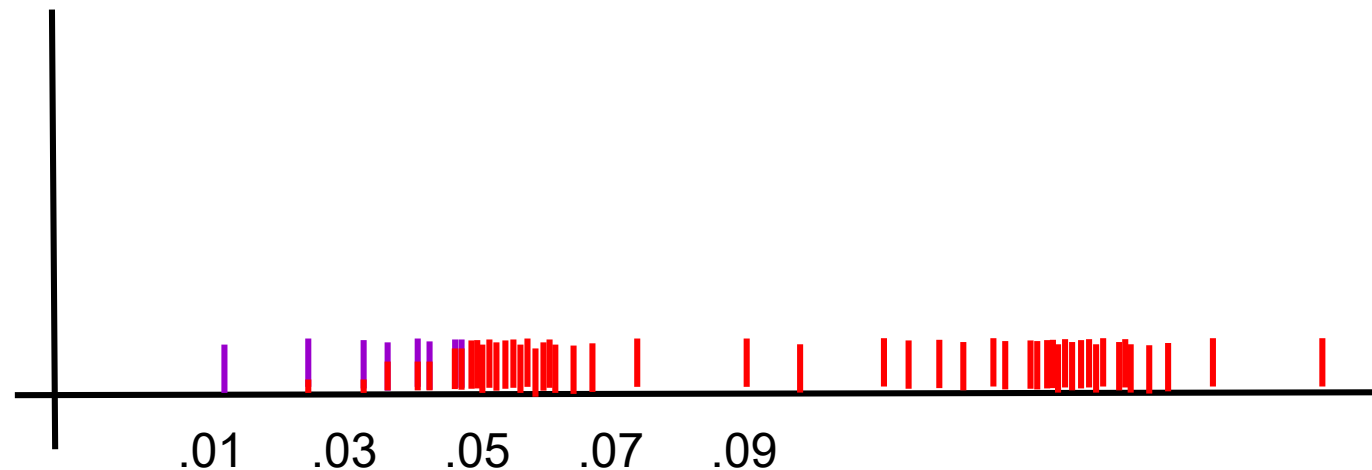
Expectation Maximization (EM)

- [E step] Compute probability of each instance having each possible label



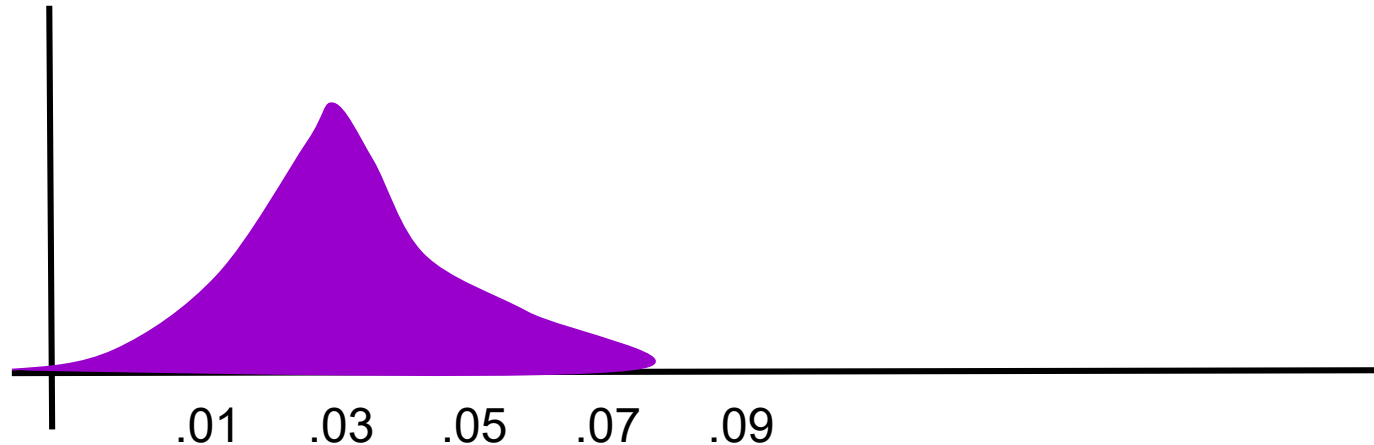
Expectation Maximization (EM)

- **[E step]** Compute probability of each instance having each possible label
- **[M step]** Treating each instance as fractionally having both labels, compute the new parameter values



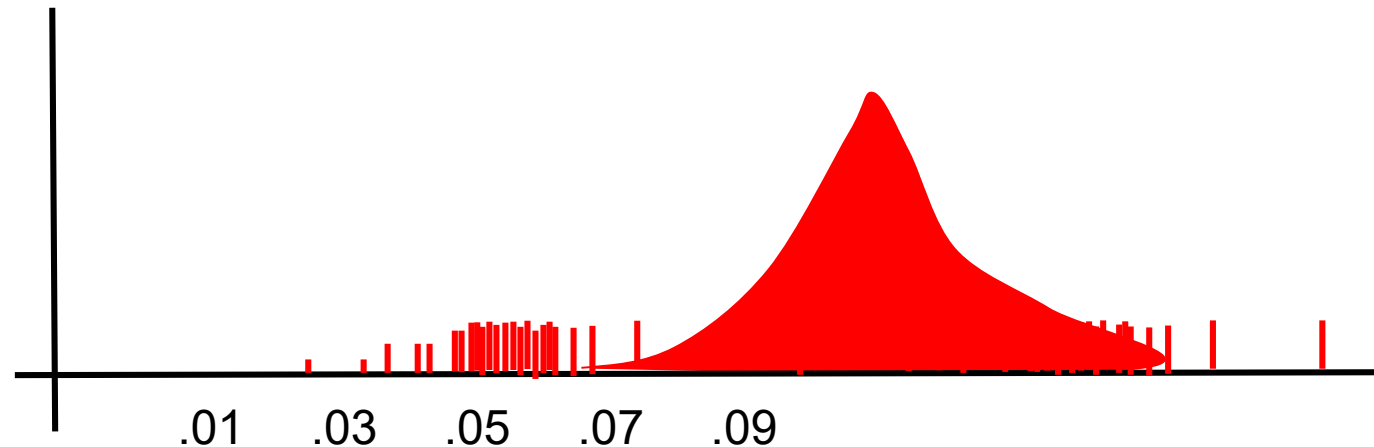
Expectation Maximization (EM)

- **[E step]** Compute probability of each instance having each possible label
- **[M step]** Treating each instance as fractionally having both labels, compute the new parameter values



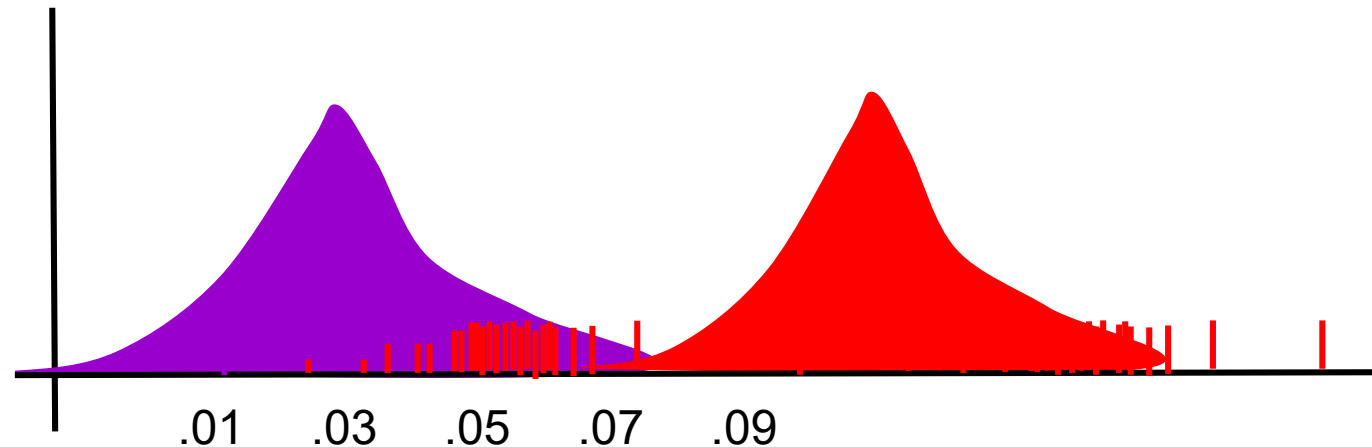
Expectation Maximization (EM)

- [E step] Compute probability of each instance having each possible label
- [M step] Treating each instance as fractionally having both labels, compute the new parameter values



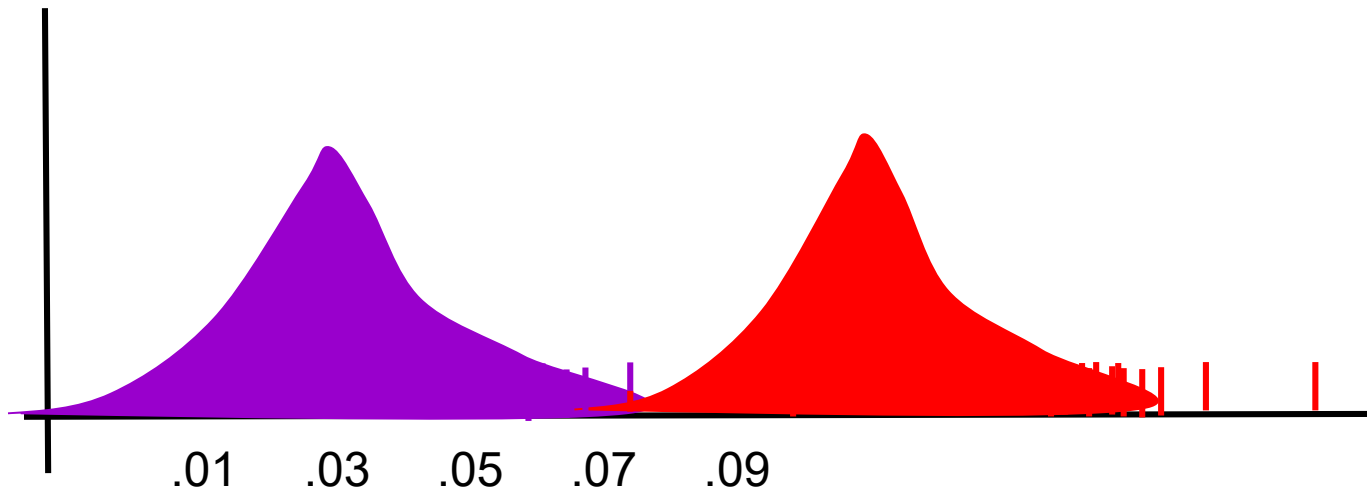
Expectation Maximization (EM)

- **[E step]** Compute probability of each instance having each possible label
- **[M step]** Treating each instance as fractionally having both labels, compute the new parameter values



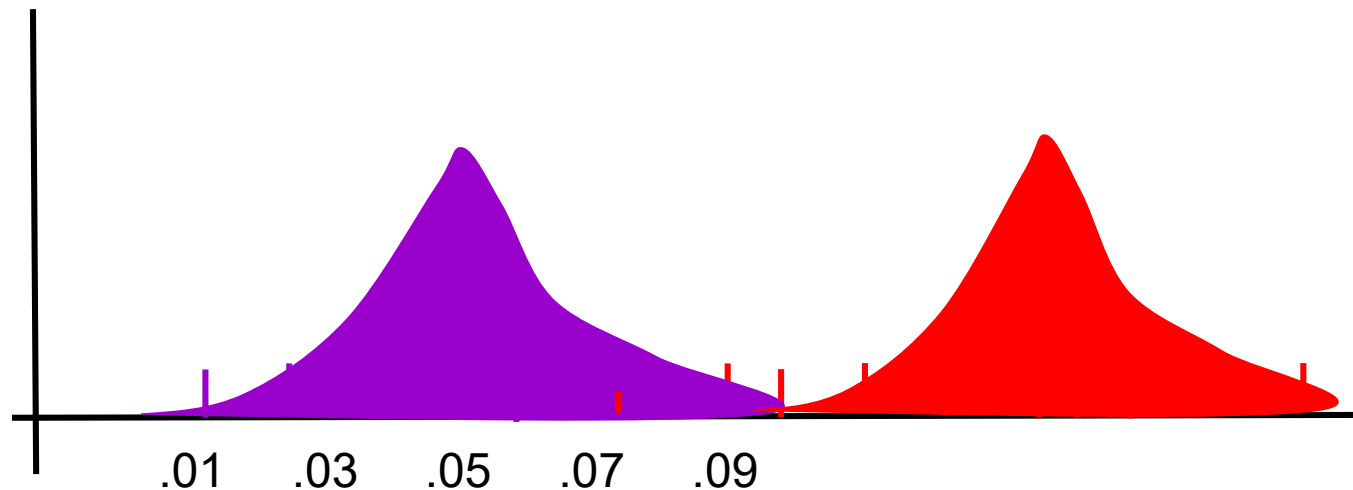
Expectation Maximization (EM)

- Repeat E-step



Expectation Maximization (EM)

- Repeat E-step
- Repeat M-step
- ... until convergence



Expectation Maximization (EM)

- Pick K random cluster models (Gaussians)
- Alternate:
 - Assign data instances **proportionately** to different models
 - Revise each cluster model based on its (**proportionately**) assigned points
- Stop when no changes

EM: Two Easy Steps

Objective: $\operatorname{argmax}_{\theta} \prod_j \sum_{i=1}^k P(y_j=i, x_j | \theta) = \sum_j \log \sum_{i=1}^k P(y_j=i, x_j | \theta)$

Data: $\{x_j \mid j=1 \dots n\}$

Notation a bit
inconsistent
Parameters = $\theta = \lambda$

- **E-step:** Compute expectations to “fill in” missing y values according to current parameters, θ
 - For all examples j and values i for y , compute: $P(y_j=i \mid x_j, \theta)$
- **M-step:** Re-estimate the parameters with “weighted” MLE estimates
 - Set $\theta = \operatorname{argmax}_{\theta} \sum_j \sum_{i=1}^k P(y_j=i \mid x_j, \theta) \log P(y_j=i, x_j | \theta)$

Especially useful when the E and M steps have closed form solutions!!!

EM for GMM

Iterate: On the t 'th iteration let our estimates be

$$\theta^{(t)} = \{ \mu_1^{(t)}, \mu_2^{(t)} \dots \mu_k^{(t)}, \Sigma_1^{(t)}, \Sigma_2^{(t)} \dots \Sigma_k^{(t)}, \pi_1^{(t)}, \pi_2^{(t)} \dots \pi_k^{(t)} \}$$

E-step

Compute label distribution of each data point

$$P(y_j = i | x_j, \theta^{(t)}) \propto \pi_i^{(t)} N(x_j | \mu_i^{(t)}, \Sigma_i^{(t)})$$

Just evaluate a Gaussian at x_j

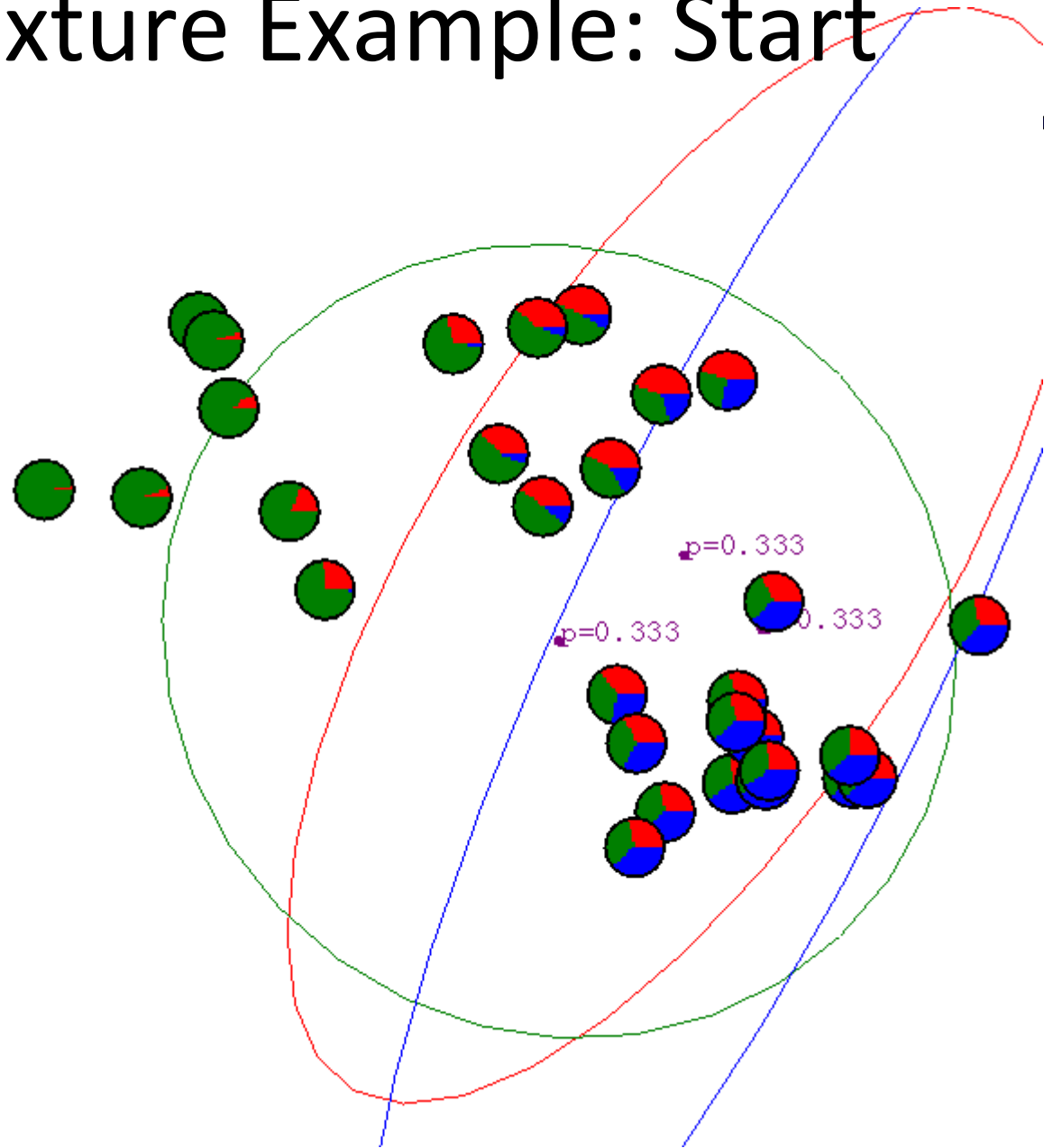
M-step

Compute weighted MLE of parameters given label distributions

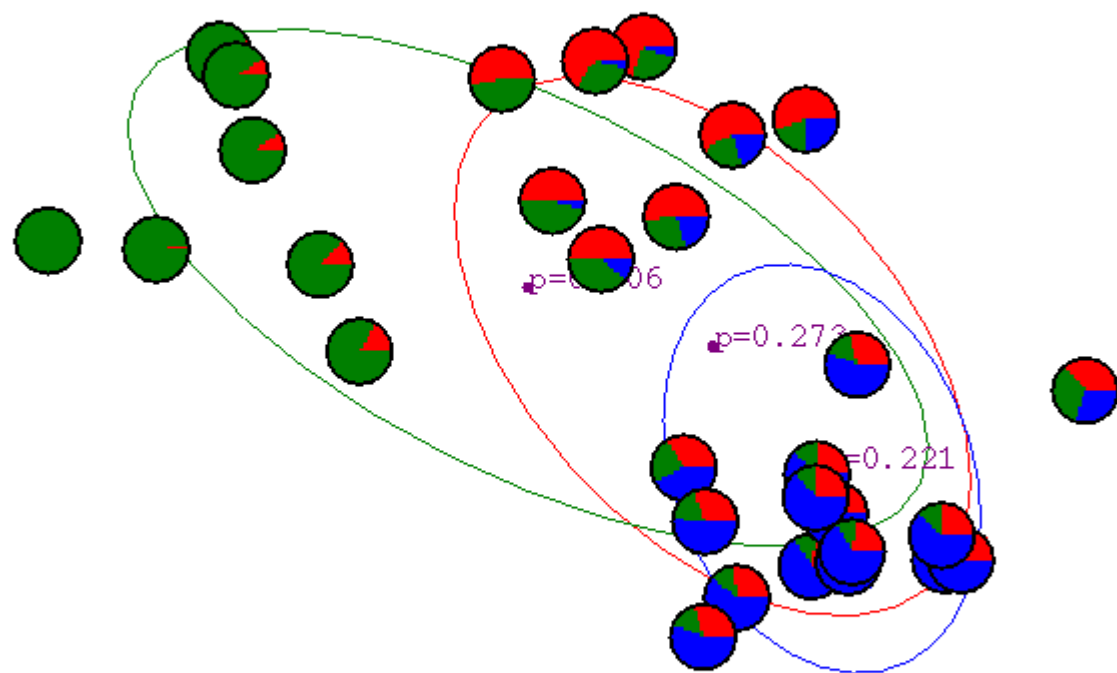
$$\mu_i^{(t+1)} = \frac{\sum_j P(y_j = i | x_j, \theta^{(t)}) x_j}{\sum_{j'} P(y_{j'} = i | x_{j'}, \theta^{(t)})} \quad \Sigma_i^{(t+1)} = \frac{\sum_j P(y_j = i | x_j, \theta^{(t)}) [x_j - \mu_i^{(t+1)}][x_j - \mu_i^{(t+1)}]^T}{\sum_{j'} P(y_{j'} = i | x_{j'}, \theta^{(t)})} \quad \pi_i^{(t+1)} = \frac{\sum_j P(y_j = i | x_j, \theta^{(t)})}{m}$$

$m = \text{\#training examples}$

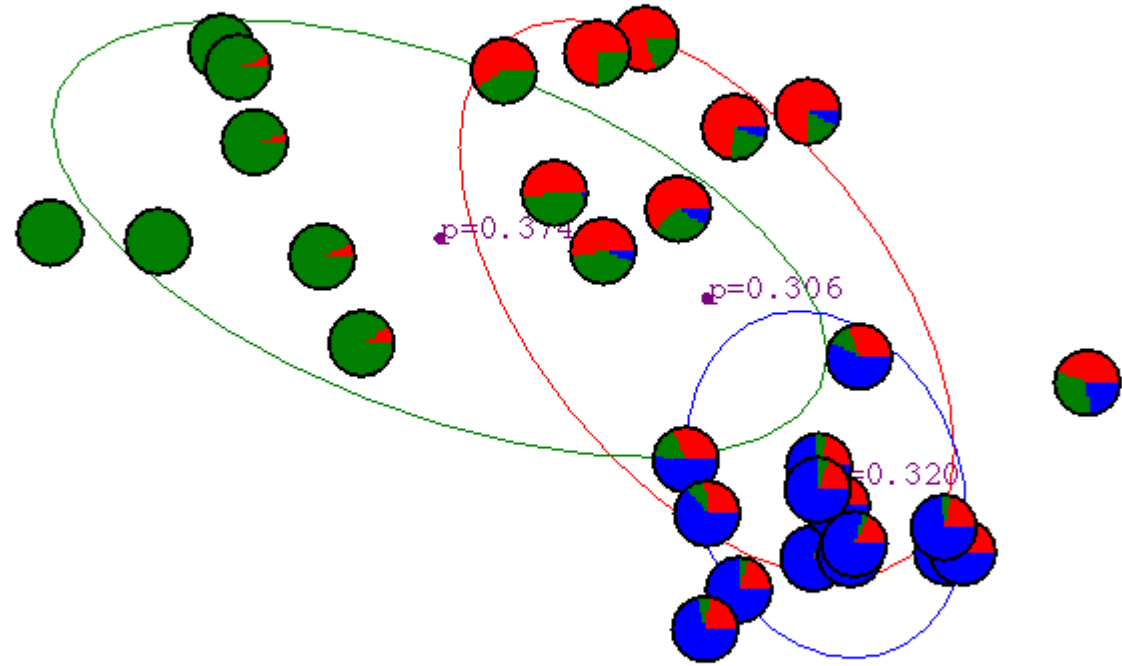
Gaussian Mixture Example: Start



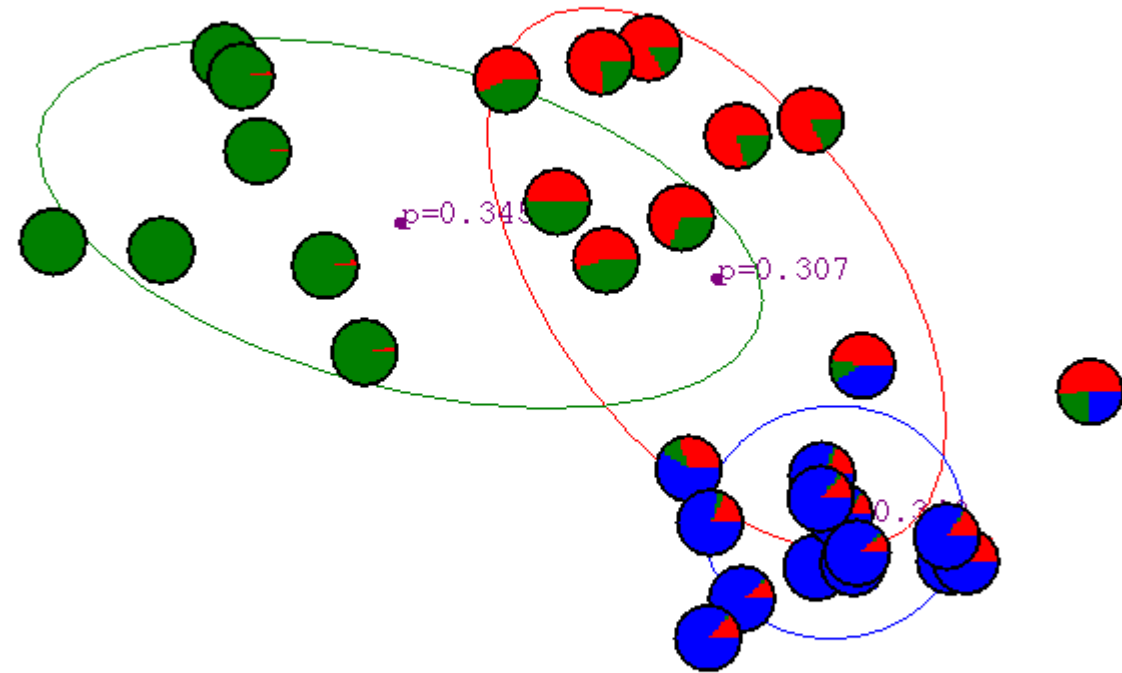
After first iteration



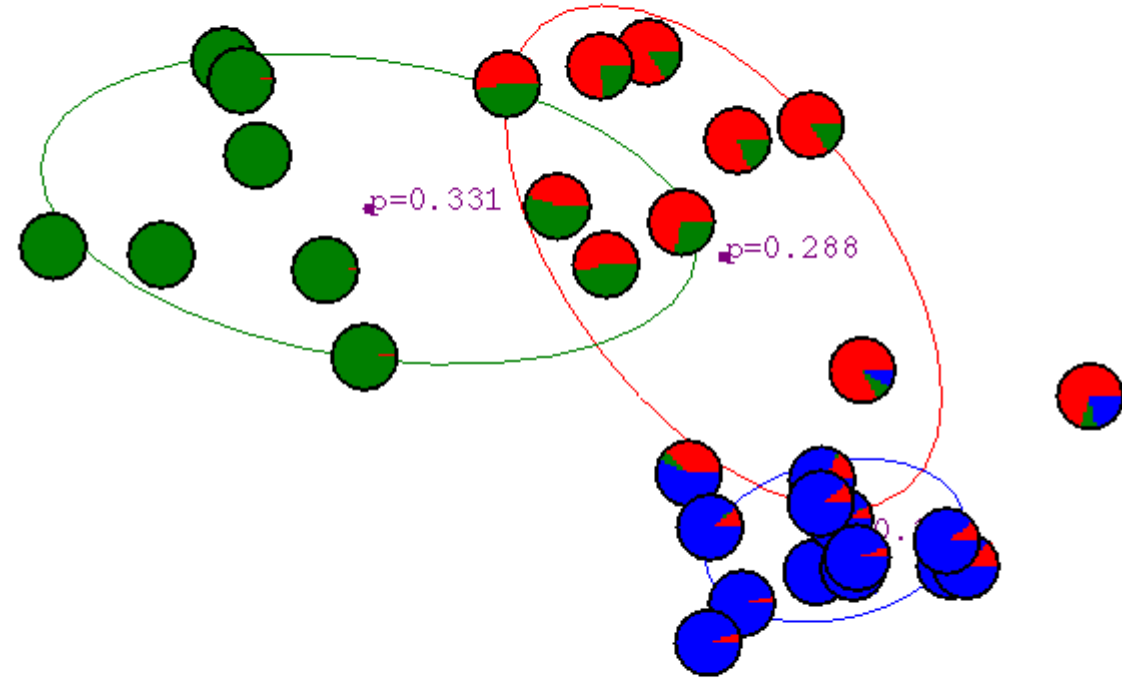
After 2nd iteration



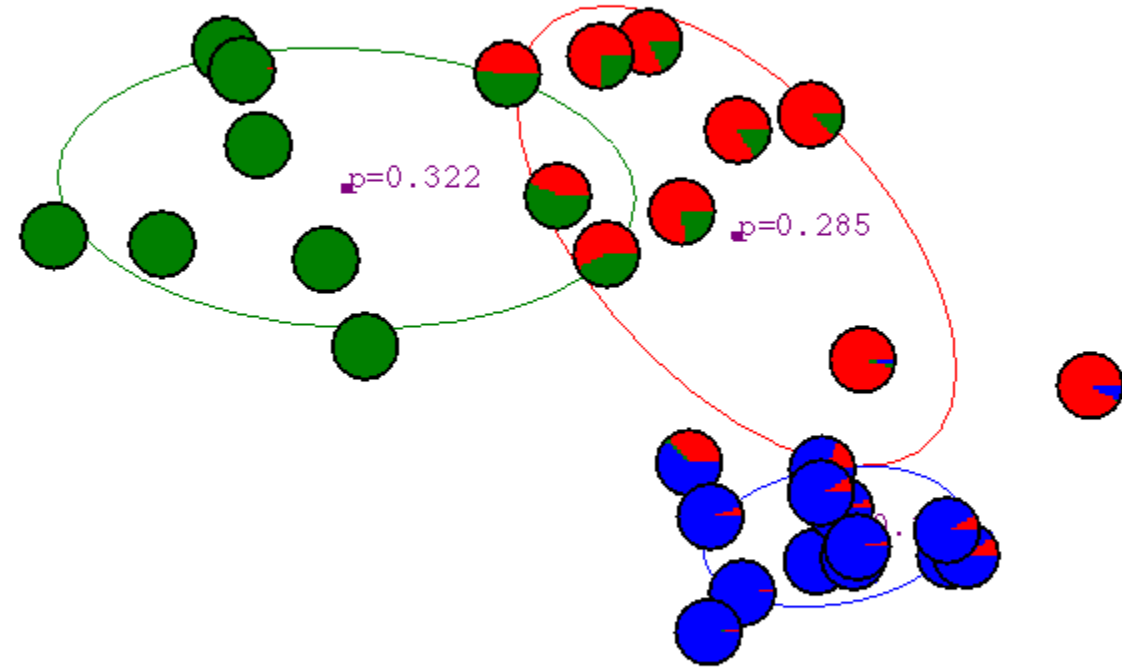
After 3rd iteration



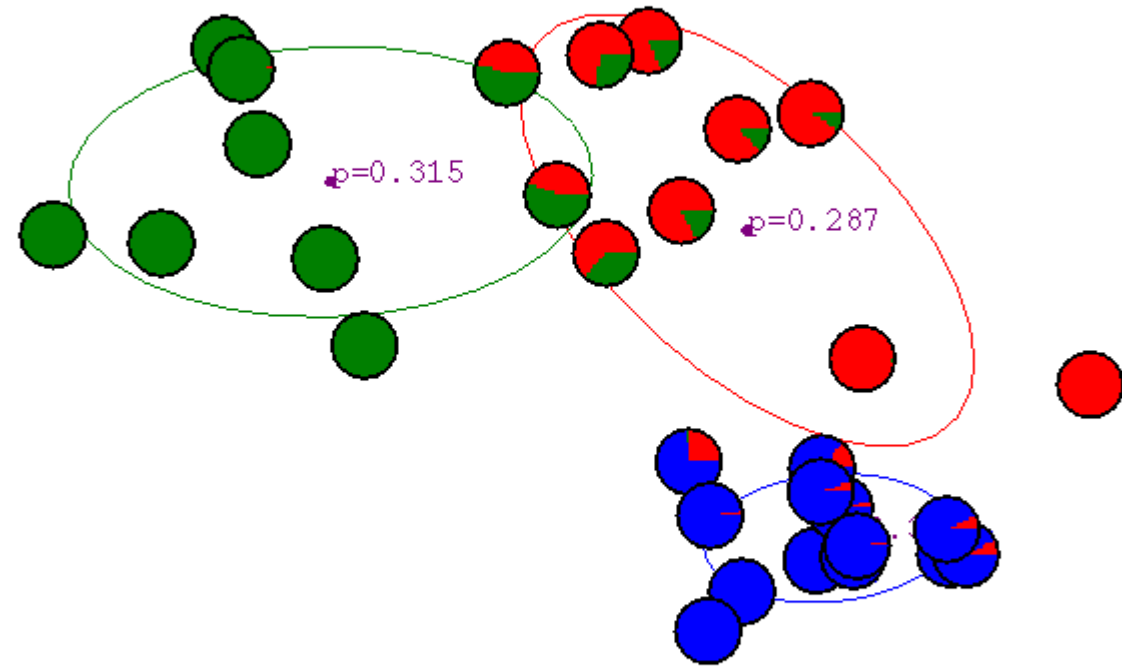
After 4th iteration



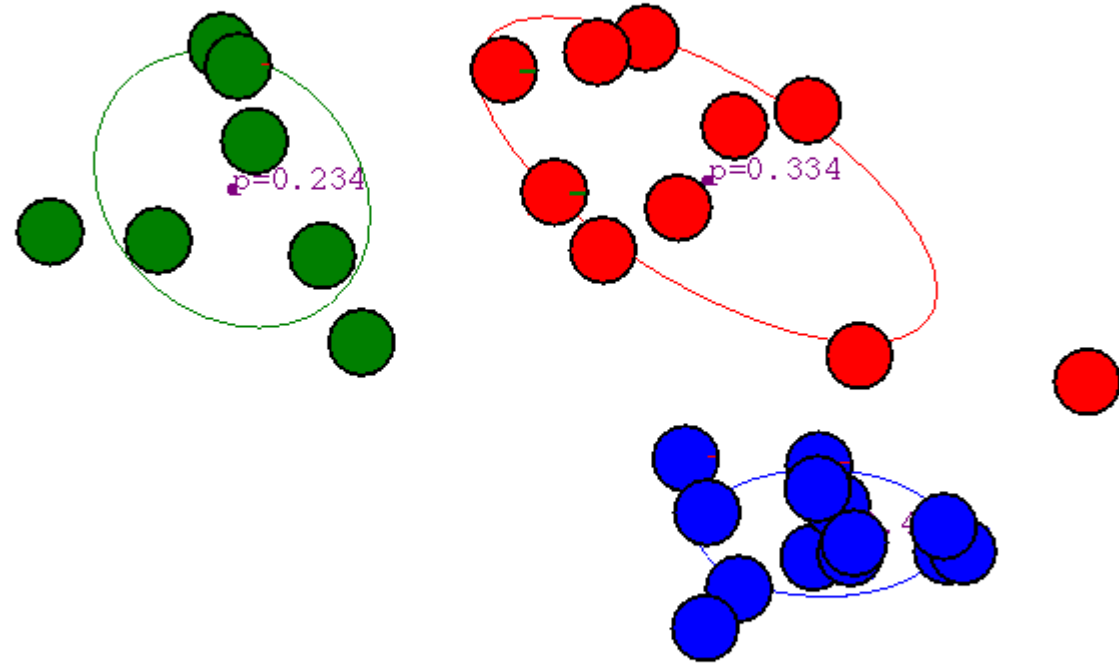
After 5th iteration



After 6th iteration



After 20th iteration



EM and K-means

- EM degrades to k-means if we assume
 - All the Gaussians are spherical and have identical weights and covariances
 - i.e., the only parameters are the means
 - The label distributions computed at E-step are point-estimations
 - i.e., hard-assignments of data points to Gaussians
 - Alternatively, assume the variances are close to zero

EM in General

- Can be used to learn any model with hidden variables (missing data)
- Alternate:
 - Compute distributions over hidden variables based on current parameter values
 - Compute new parameter values to maximize expected log likelihood based on distributions over hidden variables
- Stop when no changes

Summary

- Clustering
 - Group together similar instances
- K-means
 - Assign data instances to closest mean
 - Assign each mean to the average of its assigned points
- EM
 - Assign data instances proportionately to different Gaussian models
 - Revise each model based on its (proportionately) assigned points